

The Skin

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MORE THAN A MECHANICAL BARRIER

More than a century and a half ago, the venerable pathologist Rudolph Virchow described the skin as a mere protective covering for more delicate and functionally sophisticated internal viscera. Subsequently, the skin was viewed as a necessary but rather uninteresting barrier to fluid loss and mechanical injury. Over the last several decades, however, the skin has come to be appreciated as a surprisingly complicated organ—the largest in the body—in which precisely regulated cellular and molecular interactions govern many essential processes.

Although the human integument may appear drab compared with the skin and pelage of other members of the animal kingdom, it is extraordinarily vibrant in terms of the diversity of functions it carries out. Chief among these is its role as one of the first lines of defense against potentially harmful infectious and physical agents. The skin is not only a sophisticated sensory organ but is also important for the synthesis of vitamin D (Chapter 9), which is “powered” by sun

exposure. It is composed of several cell types and structures that function interdependently and cooperatively (Fig. 25.1).

- *Squamous epithelial cells (keratinocytes)* are normally “glued” tightly together by cell junctions known as desmosomes and produce abundant amounts of keratin protein, both of which serve to create a tough, durable physical barrier. In addition, keratinocytes secrete soluble molecules such as cytokines and defensins that augment and regulate cutaneous immune responses (described later).
- *Melanocytes* within the epidermis are responsible for the production of melanin, a brown pigment that absorbs and protects against potentially injurious ultraviolet (UV) radiation in sunlight. Melanin is subsequently transferred to and accumulates within keratinocytes, a process that determines skin color.
- *Dendritic cells*. Skin is one of the first lines of defense against microorganisms. It is constantly exposed to microbial and nonmicrobial antigens, which are processed by intraepidermal dendritic cells known as *Langerhans cells*. Langerhans cells secrete factors that augment innate immune responses and migrate from the skin

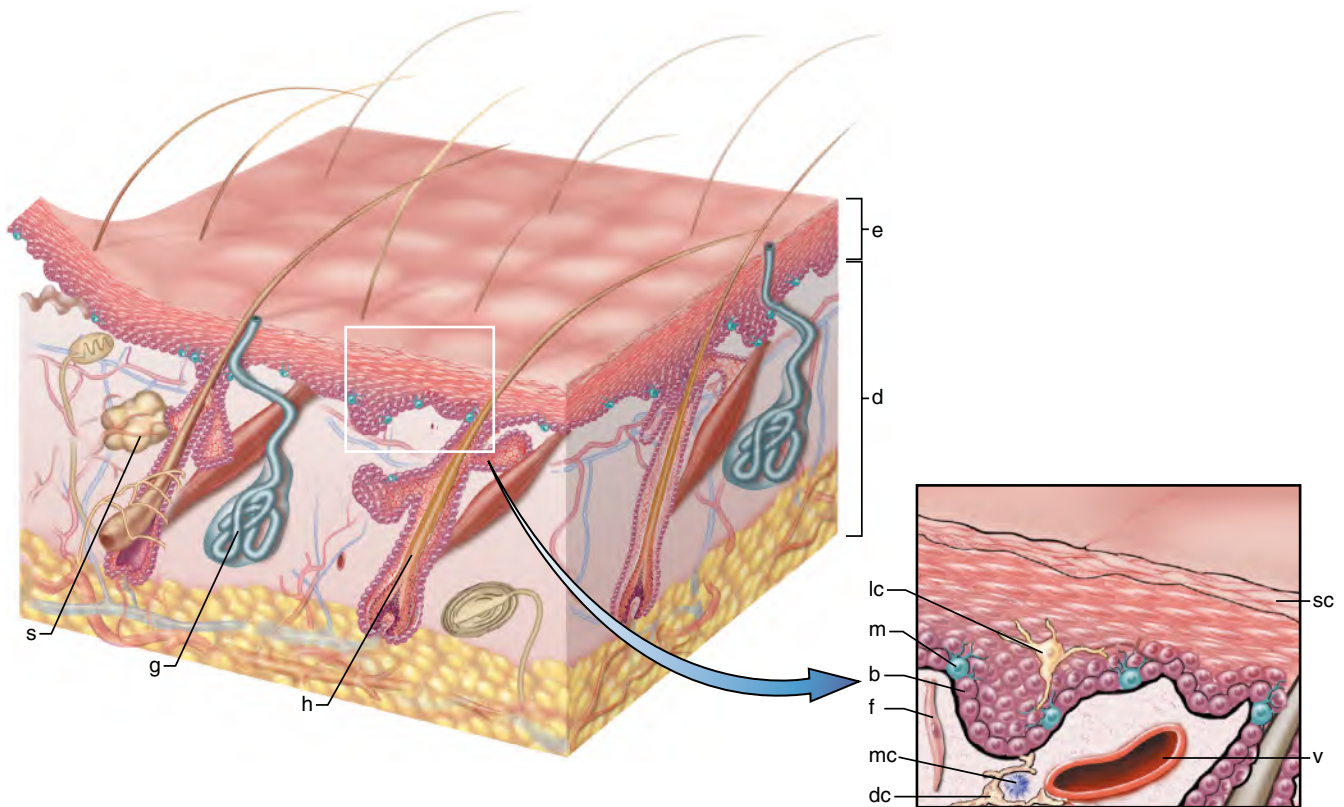


Fig. 25.1 *Left*, The skin is composed of an epidermal layer (*e*), from which specialized adnexa—hair follicles (*h*), sweat glands (*g*), and sebaceous glands (*s*)—descend into the underlying dermis (*d*). *Inset*, This projection of the epidermal layer and underlying superficial dermis demonstrates the progressive upward maturation of basal cells (*b*) into cornified squamous epithelial cells of the stratum corneum (*sc*). Melanin-containing dendritic melanocytes (*m*) (and rare Merkel cells containing neurosecretory granules) and mid-epidermal dendritic Langerhans cells (*lc*) are also present. The underlying dermis contains small vessels (*v*), fibroblasts (*f*), perivascular mast cells (*mc*), and dendrocytes (*dc*), which participate in dermal immune responses and repair.

to regional lymph nodes, where they present their antigenic cargo to T lymphocytes, thereby stimulating the adaptive immune system. Dendritic cells found within the dermis, perform similar functions.

- **Lymphocytes.** Following their stimulation by dendritic cells in regional lymph nodes, T cells expressing an adhesion molecule called cutaneous lymphocyte-associated antigen (CLA) and chemokine receptors such as CCR4 and CCR10 leave the lymph node and home back to the dermis, a process that is directed in part by chemokines secreted by activated keratinocytes. Cytokines produced by these T cells mediate the microscopic patterns and clinical features of cutaneous inflammatory and infectious diseases. In addition, small numbers of B cells are found in the dermis that can participate in humoral responses to antigens encountered in the skin.
- *The skin is a large and complex ecosystem that provides niches for a variety of organisms, including viruses, bacteria, fungi, and mites.* These organisms have evolved symbiotic relationships with their human hosts and appear to contribute to health in several ways. By occupying skin niches, the normal “zoo” of organisms (the skin microbiome) prevents colonization of the skin by other potentially harmful organisms, just as the gut microbiome protects against gastrointestinal pathogens (Chapter 17).
- *Afferent nerve fibers and a diverse set of associated specialized structures referred to as neural end organs* are responsible for physical sensations that run the gamut from pleasurable to painful, including

touch, vibration, itchiness, cold, and heat. In addition, autonomic efferent nerve fibers regulate adnexal components such as sweat glands and effector pili muscles and release factors that influence the function of innate and adaptive immune cells in the dermis. Another cell type found in skin that remains cloaked in mystery is the *Merkel cell*; these cells are found in the epithelial basal cell layer and may have neuroendocrine or mechanoreceptor functions.

- **Adnexal components.** These include sweat and sebaceous glands and hair follicles. Sweat glands help to regulate body temperature, and sebaceous glands secrete sebum, which lubricates and protects the skin, helps prevent water loss, and has antimicrobial properties. In addition to manufacturing hair shafts, hair follicles have protected niches harboring stem cells capable of regenerating superficial epithelial skin structures following their disruption by trauma, burns, and other types of injuries.

Perturbations that disrupt the delicate homeostasis that exists among skin cells may produce conditions as varied as wrinkles and hair loss, blisters and rashes, life-threatening cancers, and disorders of immune regulation. For example, long-term exposure to sunlight fosters premature cutaneous aging, blunts immunologic responses to environmental antigens, and induces the development of several malignant cutaneous neoplasms. Ingested agents, such as therapeutic drugs, can cause an enormous number of rashes or exanthems. Systemic disorders, such as diabetes mellitus, amyloidosis, and lupus erythematosus, can also have important manifestations in the skin.

TABLE 25.1 Nomenclature of Skin Lesions

Definition	
Macroscopic Lesions	
Excoriation	Traumatic lesion breaking the epidermis and causing a raw linear defect (i.e., deep scratch); often self-induced.
Lichenification	Thickened, rough skin (similar to lichen on a rock); usually the result of repeated rubbing.
Macule, Patch	Circumscribed, flat lesion distinguished from surrounding skin by color. Macules are ≤ 10 mm in diameter; patches are > 10 mm.
Onycholysis	Separation of nail plate from nail bed.
Papule, Nodule	Elevated dome-shaped or flat-topped lesion. Papules are ≤ 10 mm across; nodules are > 10 mm in size.
Plaque	Elevated flat-topped lesion, usually > 10 mm across (may be caused by coalescent papules).
Pustule	Discrete, pus-filled, raised lesion.
Scale	Dry, horny, plate-like excrescence; usually the result of imperfect cornification.
Vesicle, Bulla, Blister	Fluid-filled raised lesion 10 mm or less across (vesicle) or > 10 mm across (bulla). Blister is the common term for either lesion.
Wheal	Itchy, transient, elevated lesion with variable blanching and erythema formed as the result of dermal edema.
Microscopic Lesions	
Acanthosis	Diffuse epidermal hyperplasia.
Dyskeratosis	Abnormal, premature keratinization within cells below the stratum granulosum.
Erosion	Discontinuity of the skin showing incomplete loss of the epidermis.
Exocytosis	Infiltration of the epidermis by inflammatory cells.
Hydropic swelling (ballooning)	Intracellular edema of keratinocytes, often seen in viral infections.
Hypergranulosis	Hyperplasia of the stratum granulosum, often due to intense rubbing.
Hyperkeratosis	Thickening of the stratum corneum, often associated with a qualitative abnormality of the keratin.
Lentiginous	Linear pattern of melanocyte proliferation within the epidermal basal cell layer.
Papillomatosis	Surface elevation caused by hyperplasia and enlargement of contiguous dermal papillae.
Parakeratosis	Keratinization with retained nuclei in the stratum corneum. On mucous membranes, parakeratosis is normal.
Spongiosis	Intercellular edema of the epidermis.
Ulceration	Discontinuity of the skin marked by complete loss of the epidermis revealing dermis or subcutis.
Vacuolization	Formation of vacuoles within or adjacent to cells; often refers to basal cell–basement membrane zone area.

Skin conditions are very common, affecting about one-third of the US population each year, and the experienced observer can glean numerous insights into the functional state of the body by examining the skin. Close attention to the appearance and distribution of skin lesions is critical, as these characteristics help in establishing diagnoses and understanding pathogenesis. To underscore this point, special emphasis is placed on the gross appearance of skin lesions under each specific entity.

Thousands of diseases affect the skin. Only a few of these that are common or illustrate important pathologic mechanisms are described here. Specific terms describe the gross and microscopic appearance of skin lesions; every student must be familiar with these to be fluent in dermatopathology. The most important of these terms and their respective definitions are listed in [Table 25.1](#).

DISORDERS OF PIGMENTATION AND MELANOCYTES

Focal or widespread loss of normal protective pigmentation can make individuals extraordinarily vulnerable to the harmful effects of sunlight (as in albinism). Changes in preexisting skin pigmentation may signify important primary skin disorders (e.g., malignant transformation of a mole) or point to the existence of an underlying systemic disorder (e.g., Addison disease; see Chapter 24).

Freckle (Ephelis)

Freckles are the most common pigmented lesions of childhood in lightly pigmented individuals. It is unclear whether freckles result from a focal abnormality in pigment production in a discrete field of melanocytes, enhanced melanin transfer to adjacent basal keratinocytes, or some combination thereof. The *café au lait* spots seen in neurofibromatosis type 1 (Chapter 27) resemble freckles histologically, but are larger, arise independently of sun exposure, and contain aggregated melanosomes (macromelanosomes), which can be seen within the cytoplasm of melanocytes in electron micrographs.

MORPHOLOGY

Freckles are generally small (1 mm to several millimeters in diameter), tan-red or light brown macules that appear after sun exposure. Hyperpigmentation of freckles results from increased amounts of melanin pigment within basal keratinocytes. Associated melanocytes may be slightly enlarged but are normal in density. Once present, freckles fade and darken in a cyclic fashion during winter and summer, respectively. This is not due to changes in the number of melanocytes, but rather in the degree of pigmentation.

Lentigo

The term *lentigo* (plural, *lentigines*) refers to a common benign localized hyperplasia of melanocytes. They are equally common in both

sexes and in people with lightly and darkly pigmented skin. They may occur at any age but are most common in infants and children.

MORPHOLOGY

Lentigines may involve mucous membranes as well as the skin and consist of small (5 to 10 mm across), oval, tan-brown macules or patches. The essential histologic feature is **linear (nonnested) melanocytic hyperplasia** restricted to the cell layer immediately above the basement membrane that produces a hyperpigmented basal cell layer. Unlike freckles, lentigines do not darken when exposed to sunlight.

Melanocytic Nevus (Pigmented Nevus, Mole)

Melanocytic nevi (known colloquially as moles) are common benign neoplasms strongly associated with acquired activating mutations in components of the RAS signaling pathway. Most of us have at least a few “moles” and probably regard them as mundane and uninteresting. However, in truth, moles (or melanocytic nevi) are diverse, dynamic, and biologically fascinating. There are numerous subtypes of melanocytic nevi that are distinguished based on their clinical and histologic features; [Table 25.2](#) provides a summary of salient features of some commonly encountered forms. Acquired melanocytic nevi are the most common type and are found in virtually all individuals.

Pathogenesis

Evidence that nevi are neoplasms comes from studies showing that many have acquired mutations that lead to constitutive activation of RAS or the serine/threonine kinase BRAF, which lies immediately downstream of RAS (described in Chapter 7 and later under Melanoma). Given that RAS signals have potent transforming activity and have key roles in many full-blown cancers, it is reasonable to ask why nevi only rarely give rise to melanomas. One answer appears to lie in the phenomenon of *oncogene-induced senescence*. Expression of either activated RAS or BRAF in normal human melanocytes causes only a limited period of proliferation that is followed by a permanent growth arrest mediated by the accumulation of p16, a potent inhibitor of several cyclin-dependent kinases, including CDK4 and CDK6 (Chapter 7). This protective response is disrupted in melanoma and some precursor lesions that give rise to melanoma.

MORPHOLOGY

Common acquired melanocytic nevi are tan to brown, uniformly pigmented, small (usually less than 6 mm across), relatively flat macules or elevated papules with well-defined, rounded borders ([Figs. 25.2A and 25.3A](#)). They may become more prominent during pregnancy, indicating a degree of hormone sensitivity. Melanocytic nevi are thought to progress through a series of morphologic changes over time. The earliest lesions are believed to be **junctional nevi**, which consist of aggregates or nests of round cells that grow along the dermoepidermal junction ([Fig. 25.2B](#)). Nuclei of nevus cells are uniform and rounded in contour, contain inconspicuous nucleoli, and show little or no mitotic activity. Eventually, most junctional nevi grow into the underlying dermis as nests or cords of cells to form **compound nevi** ([Fig. 25.3B](#)). In older lesions the epidermal nests may be lost entirely to form pure **intra-dermal nevi**. Compound and dermal nevi are often more elevated than junctional nevi.

Growth of nevus cells from the dermoepidermal junction into the underlying dermis is accompanied by morphologic changes that are believed to reflect oncogene-induced senescence ([Fig. 25.4](#)). Whereas superficial nevus cells are larger, tend to produce melanin, and grow in nests, deeper nevus cells are smaller, produce little or no pigment, and appear as cords and single cells. At the deepest extent of the lesions, these cells can acquire fusiform contours and grow in fascicles resembling neural tissue (**neurotization**; see [Fig. 25.4E](#)). This striking metamorphosis correlates with enzymatic changes (progressive loss of tyrosinase activity) in deeper, nonpigmented, “nerve-like” nevus cells. These changes are helpful in distinguishing nevi from melanomas, which lack such features.

Although melanocytic nevi are common, their clinical and histologic diversity necessitates thorough knowledge of their appearance and natural history, lest they be confused with other skin conditions, most notably melanoma. In addition, dysplastic nevi are important to recognize because they are associated with an increased risk of melanoma (described next).

Dysplastic Nevus

Dysplastic nevi can rarely be direct precursors of melanoma and when multiple in number are a marker of an increased risk for melanoma. The association of melanocytic nevi with melanoma was made almost 200 years ago, but a potential precursor of melanoma was not identified until 1978 when Clark and colleagues described lesions now referred to as *dysplastic nevi*. Several lines of evidence support the concept that some dysplastic nevi are precursors of melanoma. One of

TABLE 25.2 Representative Variant Forms of Melanocytic Nevus

Nevus Variant	Diagnostic Architectural Features	Cytologic Features	Clinical Significance
Congenital nevus	Deep dermal and sometimes subcutaneous growth around adnexa, neurovascular bundles, and blood vessel walls	Identical to ordinary acquired nevi	Present at birth; large variants have increased melanoma risk
Blue nevus	Nonnested dermal infiltration, often with associated fibrosis	Highly dendritic, heavily pigmented nevus cells	Black-blue nodule; often confused with melanoma clinically
Spindle and epithelioid cell nevus (Spitz nevus)	Fascicular growth	Large, plump cells with pink-blue cytoplasm; fusiform cells	Common in children; red-pink nodule; often confused with hemangioma clinically
Halo nevus	Lymphocytic infiltration surrounding nevus cells	Identical to ordinary acquired nevi	Host immune response against nevus cells and surrounding normal melanocytes
Dysplastic nevus	Coalescent intraepidermal nests	Cytologic atypia	Potential marker or precursor of melanoma

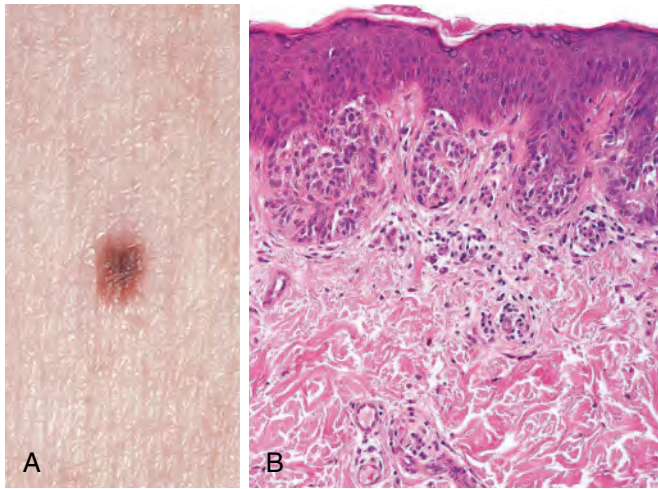


Fig. 25.2 Melanocytic nevus, junctional type. (A) Grossly, lesions are small, relatively flat, symmetric, and uniform. (B) On histologic examination, junctional nevi are characterized by rounded nests of nevus cells originating at the tips of rete ridges along the dermoepidermal junction.

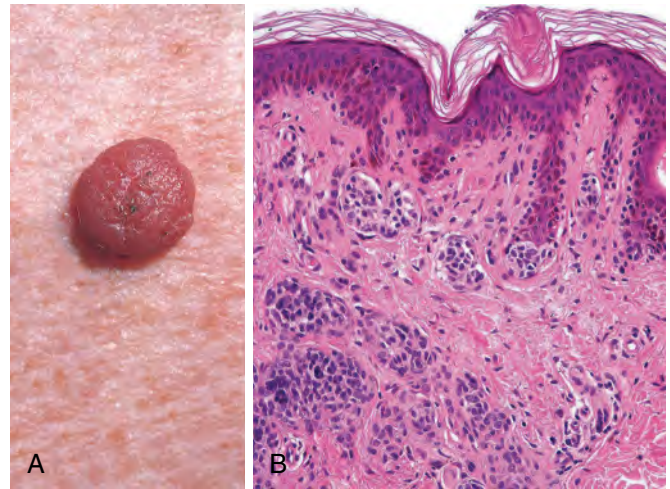


Fig. 25.3 Melanocytic nevus, compound type. (A) In contrast to the junctional nevus, the compound nevus is raised and dome shaped. The symmetry and uniform pigment distribution suggest a benign process. (B) Histologically, compound nevi combine the features of junctional nevi (intraepidermal nevus cell nests) with nests and cords of dermal nevus cells.

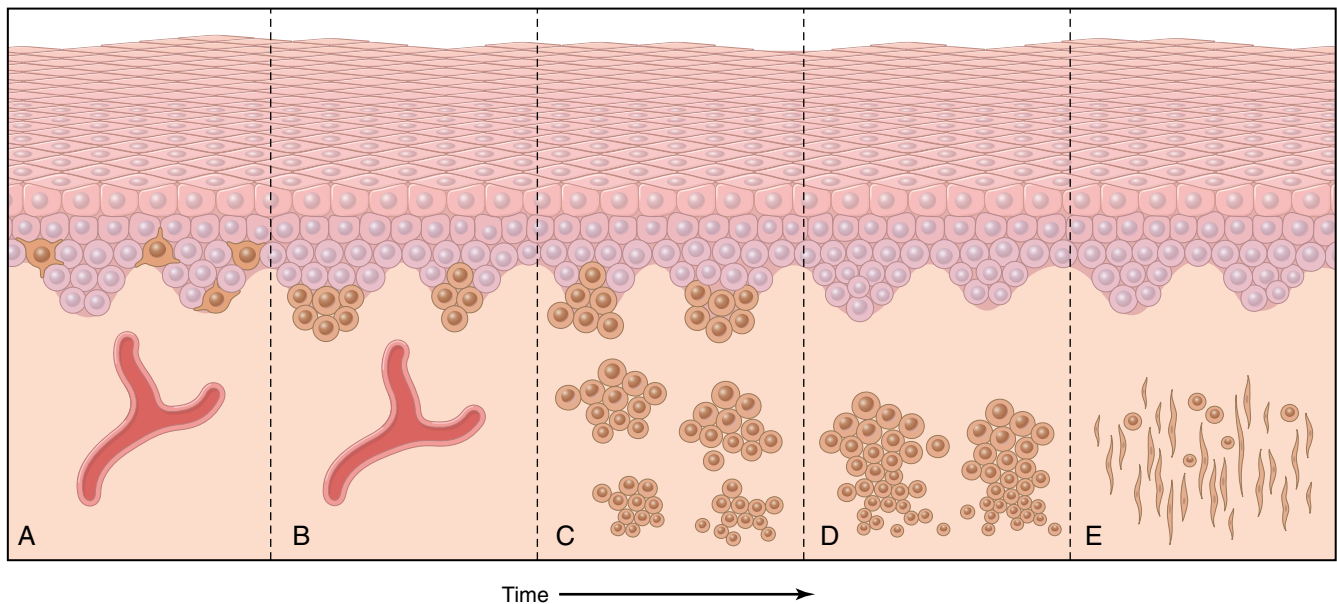


Fig. 25.4 Maturation sequence of nondysplastic melanocytic nevi. (A) Normal skin shows scattered dendritic melanocytes within the epidermal basal cell layer. (B) Junctional nevus. (C) Compound nevus. (D) Dermal nevus. (E) Dermal nevus with neurotization, a change that is also referred to as maturation. Nevi may exist at any stage in this sequence for variable periods of time, although many are believed to progress through this sequence.

the most compelling involves studies of families affected by *dysplastic nevus syndrome* (also known as familial atypical multiple mole-melanoma syndrome), an autosomal dominant disorder in which a tendency to develop multiple dysplastic nevi and melanoma are coinherited. The probability that a person with dysplastic nevus syndrome will develop melanoma is over 50% by age 60, and at-risk individuals sometimes develop melanomas at multiple sites. Even more directly, apparent transformation of dysplastic nevi to melanoma has been documented histologically.

Although they can give rise to melanoma, most dysplastic nevi are clinically stable and never progress. Conversely, not all melanomas in

individuals with dysplastic nevus syndrome arise from dysplastic nevi, suggesting that these lesions are probably best viewed as indicators of increased melanoma risk. Dysplastic nevi also occur as isolated lesions in otherwise normal individuals, in which case the risk of malignant transformation is very low.

Pathogenesis

Clark and associates first proposed morphologic stages in the development of dysplastic nevi and their progression to melanoma (Fig. 25.5), and subsequent workers correlated these morphologic changes with the stepwise acquisition of oncogenic mutations and epigenetic

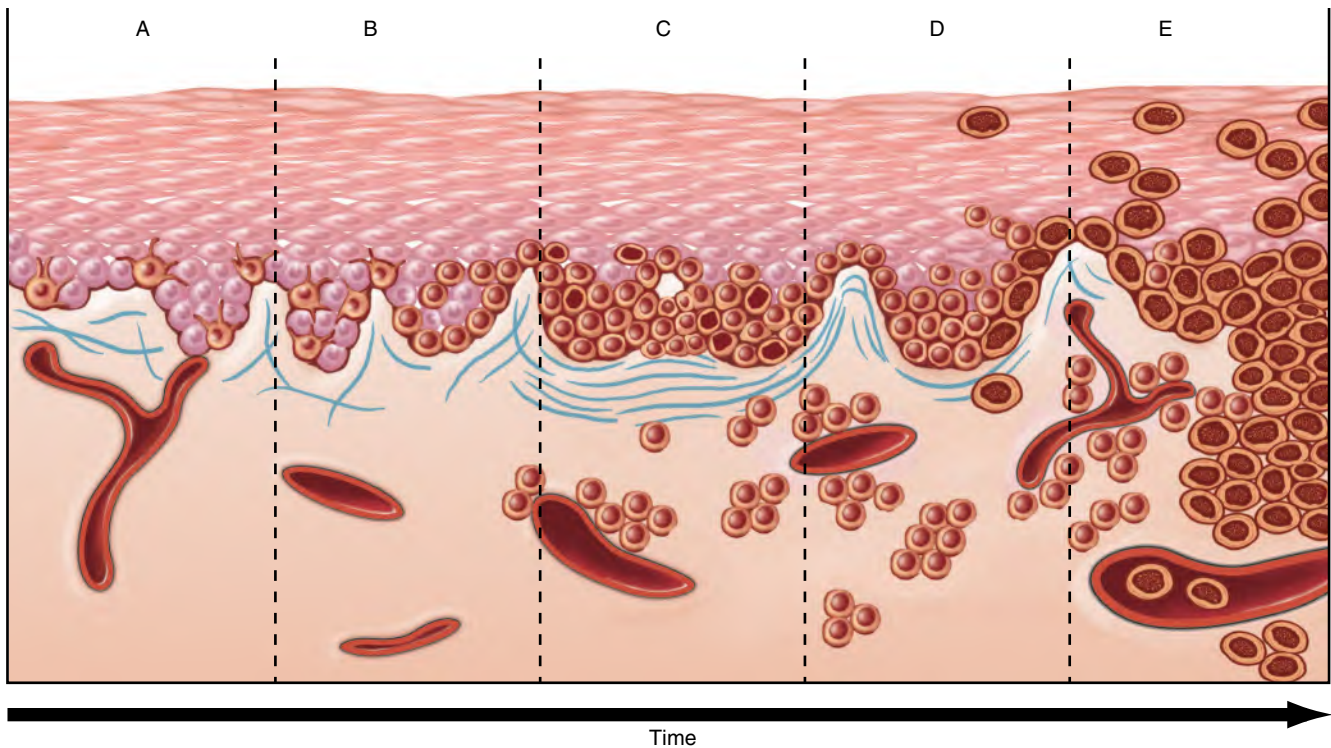


Fig. 25.5 Potential steps of tumor progression in dysplastic nevi. (A) Lentiginous melanocytic hyperplasia. (B) Lentiginous junctional nevus. (C) Lentiginous compound nevus with abnormal architectural and cytologic features (dysplastic nevus). (D) Early melanoma, or melanoma in radial growth phase (large dark cells in epidermis). (E) Advanced melanoma (vertical growth phase) with malignant spread into the dermis and vessels. The risk of malignant transformation of any single dysplastic nevus is small but appears to be higher than that of typical nevi.

changes. Like conventional nevi, dysplastic nevi also frequently have acquired activating mutations in the *NRAS* and *BRAF* genes. What then distinguishes dysplastic nevi from typical melanocytic nevi? An important clue comes from individuals with dysplastic nevus syndrome. Such individuals often have inherited loss-of-function mutations in *CDKN2A*. *CDKN2A* encodes several proteins including p16 (described in more detail under Melanoma), which you will recall is a negative regulator of both cyclin-dependent kinase 4 (CDK4) and CDK6 (Chapter 1). Other affected families have mutations in the *CDK4* gene that make the CDK4 protein resistant to inhibition by p16. Thus, it appears that RAS or BRAF activation and increased CDK4 activity contribute to the development of dysplastic nevi. However, not all patients with germline mutations in *CDKN2A* or *CDK4* have dysplastic nevi, and not all familial dysplastic nevi are associated with mutations in these genes. As a result, it is suspected that additional modifier genes influence whether dysplastic nevi occur in a particular individual.

MORPHOLOGY

Dysplastic nevi are larger than most acquired nevi (often greater than 5 mm across) and in individuals with the dysplastic nevus syndrome may number in the hundreds (Fig. 25.6A). They may appear as flat macules, slightly raised plaques with a “pebbly” surface, or target-like lesions with a darker raised center and irregular flat periphery. They can be recognized based on their large size, variability in pigmentation (variegation), and irregular borders. Most seem to be acquired rather than congenital. Unlike ordinary moles, dysplastic nevi occur on both sun-exposed and protected body surfaces.

Microscopically, dysplastic nevi usually involve the epidermis and the dermis and exhibit architectural and cytologic atypia (Fig. 25.6A,B). Nevus cell nests within the epidermis may be enlarged and often fuse or coalesce with adjacent nests. As part of this process, single nevus cells begin to replace the normal basal cell layer along the dermoepidermal junction, producing **lentiginous hyperplasia**. Cytologic atypia takes the form of nuclear enlargement; irregular, often angulated, nuclear contours; and hyperchromasia (Fig. 25.6C). Associated alterations in the superficial dermis include lymphocytic infiltrates (usually sparse); release of melanin from dead nevus cells into the dermis (melanin incontinence), where it is phagocytosed by dermal macrophages; and a peculiar **linear fibrosis** surrounding the epidermal rete ridges that are involved by the nevus. The diagnosis is based on this constellation of features, rather than any single finding.

Melanoma

Melanoma, the deadliest skin cancer, is strongly linked to acquired mutations caused by exposure to UV radiation in sunlight. Melanoma is a relatively common neoplasm that can be cured if it is detected and treated in its earliest stages. The great preponderance of melanoma arises in the skin; other sites of origin include the oral and anogenital mucosal surfaces (i.e., oropharynx, gastrointestinal and genitourinary tracts), the esophagus, the meninges, and the uvea of the eye (Chapter 29). The following comments apply to cutaneous melanomas.

Today, due to increased public awareness of the signs of cutaneous melanoma, most are cured surgically. Nevertheless, the reported incidence of melanoma is increasing worldwide; almost 100,000 cases and about 8000 deaths are expected in the United States in 2023.

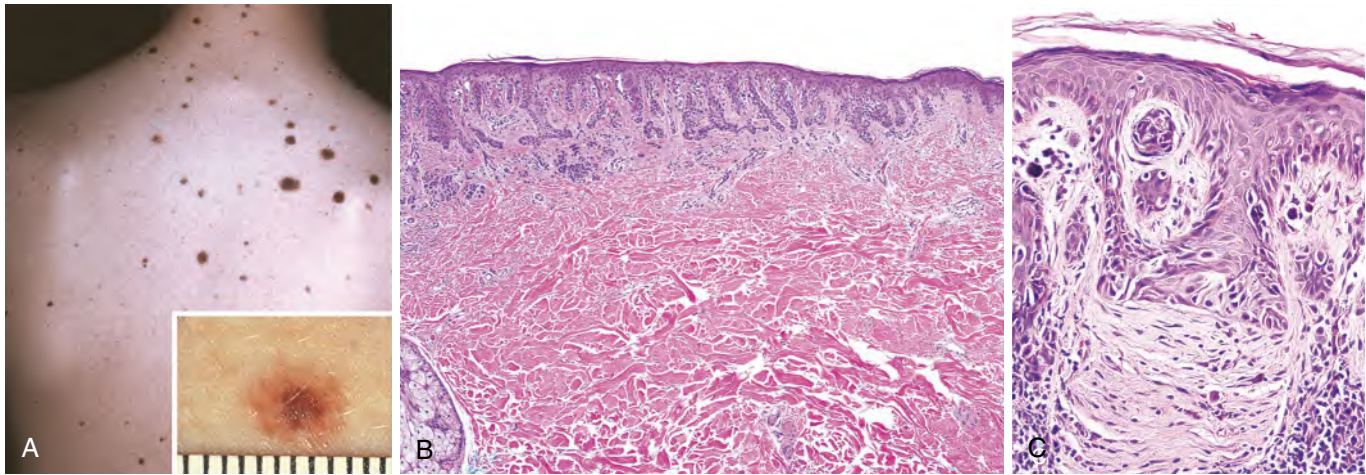


Fig. 25.6 Dysplastic nevus. (A) Numerous nevi on the back. (B) One such lesion (A, *inset*) has a compound nevus component (*left*) and an asymmetric junctional nevus component (*right*). The former corresponds to the more pigmented and raised central zone, and the latter corresponds to the less pigmented, flat peripheral rim of the lesion shown in A. (C) An important feature is the presence of cytologic atypia (irregularly shaped, dark-staining nuclei). The dermis underlying the atypical cells characteristically shows linear, or lamellar, fibrosis.

Notably, there has been a decrease in death rates over the last decade, likely reflecting the effectiveness of immune checkpoint inhibitor therapy (Chapter 7).

Pathogenesis

In up to 10% of patients, the risk of melanoma is inherited as an autosomal dominant trait with variable penetrance. As mentioned when discussing dysplastic nevi, some of these familial cases are associated with germline mutations affecting genes that regulate cell cycle progression, whereas others are associated with germline mutations affecting telomerase expression (described later). In the remaining patients, melanoma is sporadic and is strongly related to a single predisposing environmental factor: UV radiation exposure from sunlight. Because repair of UV radiation-induced DNA damage is imperfect, UV radiation leads to the accumulation of mutations in melanocytes over time. Sequencing of melanoma genomes has demonstrated a very high rate of point mutations that bear the signature of the damaging effects of UV radiation on DNA. In line with this molecular evidence, melanomas most commonly arise on sun-exposed surfaces, particularly the upper back in males and the back and legs in females, and lightly pigmented individuals are at higher risk than more darkly pigmented individuals. Other inherited genetic variants linked to an increased risk of melanoma in fair-skinned populations act by diminishing melanin production in skin, thus increasing the amount of damage that sun exposure wreaks.

Nevertheless, the relationship between sun exposure and melanoma is not as straightforward as with other skin cancers, such as squamous cell carcinoma (discussed later). Some studies suggest that periodic severe sunburns early in life are the most important risk factor. Furthermore, since melanomas sometimes occur in dark-skinned individuals and at body sites that are not sun-exposed, sunlight is not always an essential predisposing factor, and other environmental factors may also contribute to risk.

The most frequent “driver” mutations in melanoma affect cell cycle control, progrowth pathways, and telomerase. Some of the more common mutations are:

- **Mutations that disrupt cell cycle control genes.** The *CDKN2A* gene is mutated in approximately 40% of pedigrees with autosomal

dominant familial melanoma. *CDKN2A* is a complex locus that encodes two different tumor suppressors, p16 and ARF. Loss of p16 is clearly implicated in human melanoma, and experimental evidence also supports a role for loss of ARF. As already mentioned, p16 inhibits CDK4 and CDK6, thus reinforcing the ability of the RB tumor suppressor to block cells in the G_1 phase of the cell cycle. By contrast, ARF enhances the activity of the p53 tumor suppressor by inhibiting MDM2, an oncoprotein that stimulates p53 degradation. *CDKN2A* is commonly mutated in sporadic melanomas, though other mechanisms controlling the G_1 checkpoint may also be involved. The net effect of these alterations is the same: increased melanocytic proliferation due to loss of cell cycle control and escape from oncogene-induced cellular senescence.

- **Mutations that activate progrowth signaling pathways.** A second common group of molecular lesions in sporadic melanoma leads to increases in RAS and PI3K/AKT signaling (Fig. 25.7), which promote cell growth and survival (Chapter 7). Activating mutations in *BRAF*, a serine/threonine kinase that is downstream of RAS, are seen in 40% to 50% of melanomas, while activating mutations in *RAS* occur in an additional 15% to 20% of tumors. Melanomas with *BRAF* mutations often also show loss of the *PTEN* tumor-suppressor gene, leading to uncontrolled activation of the PI3K/AKT pathway. For reasons that are unclear, melanomas arising in non-sun-exposed cutaneous sites rarely have mutations in *BRAF* or *RAS* and are more likely to have activating mutations in the receptor tyrosine kinase KIT, which sits upstream of both RAS and PI3K/AKT. *PTEN* is also silenced in about 20% of melanomas arising in non-sun-exposed sites. Other melanomas have loss-of-function mutations in the tumor-suppressor gene encoding neurofibromin 1 (*NF1*), a negative regulator of RAS, yet another mechanism of unleashing RAS signaling.
- **Mutations that activate telomerase.** Reactivation of telomerase, the enzyme activity that preserves telomeres and protects cells from senescence, has long been known to be important in cancer (Chapter 7). Sequencing of sporadic melanomas has identified mutations in the promoter of *TERT*, the gene that encodes the catalytic subunit of telomerase, in roughly 70% of cases, making *TERT* the most commonly mutated gene yet identified in this cancer. Rare

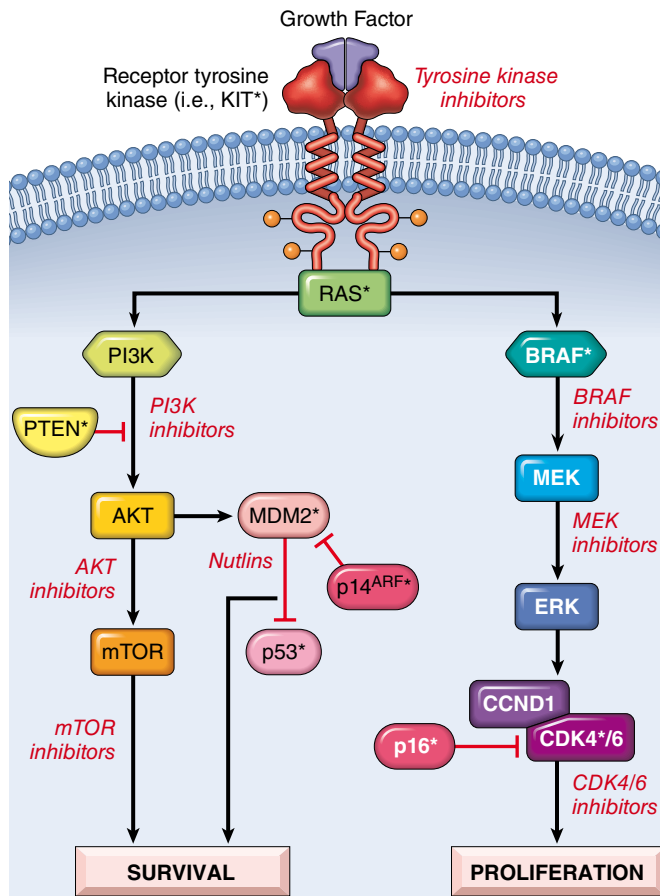


Fig. 25.7 Pathways important in melanoma. Growth factors activate signaling circuits involving receptor tyrosine kinases (e.g., *KIT*), RAS, and two key downstream pathways that include the serine/threonine kinase BRAF and the phospholipid kinase PI3K. Proteins denoted by asterisks are mutated in melanoma. Components of these pathways that are being targeted by drugs are indicated. *mTOR*, Mechanistic target of rapamycin; *PI3K*, phosphatidylinositol 3-kinase; *PTEN*, phosphatase and tensin homolog.

melanoma-prone families have also been identified that have germline *TERT* promoter mutations. As might be anticipated, the mutations increase *TERT* expression, suggesting that they act as an antidote to senescence.

MORPHOLOGY

Unlike benign nevi, melanomas show striking **variations in color**, appearing in shades of black, brown, red, dark blue, and gray (Fig. 25.8A). On occasion, zones of white or flesh-colored hypopigmentation also appear, sometimes due to focal regression of the tumor. **The borders of melanomas are irregular and often notched**, unlike the smooth, round, and uniform borders of melanocytic nevi.

Central to understanding the progression of melanoma is the concept of radial and vertical growth phases. **Radial growth** describes the horizontal spread of melanoma within the epidermis and superficial dermis (Fig. 25.8B). During this initial stage the tumor cells seem to lack the capacity to metastasize. Tumors in radial growth phase include several clinicopathologic types: **superficial spreading melanoma**, the most common type, occurs on sun-exposed skin; **lentigo maligna** usually presents as an indolent lesion on the face of older males and may remain in radial growth phase for several decades; and **acral/mucosal lentiginous melanoma**, which occurs on non-sun-exposed sites.

After a variable (and unpredictable) period, melanoma shifts from the radial phase to a **vertical growth phase**, during which the tumor cells invade downward into the dermis as an expansile mass (Fig. 25.8C). The vertical growth phase is often heralded by the appearance of a nodule and correlates with the emergence of a tumor subclone with metastatic potential. Unlike melanocytic nevi, maturation or “neurotization” is absent from the deep invasive portion of melanoma.

Individual melanoma cells are usually larger than normal melanocytes or cells found in melanocytic nevi. They have enlarged nuclei with irregular contours, chromatin that is characteristically clumped at the periphery of the nuclear membrane, and prominent red (eosinophilic) nucleoli (Fig. 25.8D). The appearance of the tumor cells is similar in the radial and vertical phases of growth. While most nevi and melanomas are easily distinguished based on their appearance, a small fraction of “atypical” lesions fall in a histologic gray zone and have been termed melanocytic tumors of uncertain malignant potential; such lesions require complete excision and close clinical follow-up.

Once a melanoma is excised, several pathologic features are used to gauge the probability of metastatic spread and prognosis. One model used to predict outcome is based in part on the (1) tumor depth of invasion (**Breslow thickness**), (2) number of mitoses, (3) evidence of tumor regression (presumably due to the host immune response), (4) ulceration of overlying skin, (5) presence and number of tumor-infiltrating lymphocytes, and (6) location (central body or extremity). Determinants of a more favorable prognosis in this model include thinner tumor depth, no or few mitoses (<1 per mm²), a brisk tumor-infiltrating lymphocyte response, absence of regression, and lack of ulceration. Since most melanomas initially metastasize to regional lymph nodes, additional prognostic information may be obtained by performing a sentinel lymph node biopsy, as in breast cancer (Chapter 23). Microscopic involvement of a sentinel node by even a small number of melanoma cells (micrometastases, Fig. 25.8D, inset) confers a worse prognosis. The size of nodal metastases and the total number of lymph nodes involved correlate with disease-specific survival.

Clinical Features

The most important warning signs, sometimes called the ABCDEs of melanoma, are (1) Asymmetry; (2) irregular Borders; (3) variegated Color; (4) increasing Diameter; and (5) Evolution or change over time, especially if rapid. Because locally advanced melanomas often metastasize, early recognition and complete excision are critical. Melanoma of the skin is usually asymptomatic, although itching or pain may be early manifestations. Most are greater than 10 mm in diameter at diagnosis. The most consistent clinical signs are changes in the color, size, or shape of a pigmented lesion.

Molecular insights into the pathogenesis of melanoma led to success in treating this cancer with drugs that target the RAS pathway (see Fig. 25.7). Such approaches were urgently needed, as metastatic melanoma is resistant to conventional chemotherapy. Recognition that melanoma is inherently immunogenic has also resulted in successful development of therapeutic regimens using antibodies that inhibit immune checkpoint proteins, such as CTLA-4 and PD-1 (programmed cell death 1). Melanoma is one of the most responsive cancers to these checkpoint inhibitors, which take the brakes off an immune system that is poised to respond to tumor neoantigens (Chapter 7).

It is currently recommended that all metastatic cutaneous melanomas be evaluated for the presence of *BRAF* mutations. Those that are *BRAF*-mutated are treated with a combination of inhibitors of *BRAF* and MEK1 (a kinase that lies immediately downstream of *BRAF*) plus immune checkpoint inhibitors. By contrast, *BRAF* wild type tumors are treated with immune checkpoint inhibitors alone. Other current efforts are focused on maximizing tumor response to checkpoint inhibitors without undue toxicity, which stems from autoimmune attack

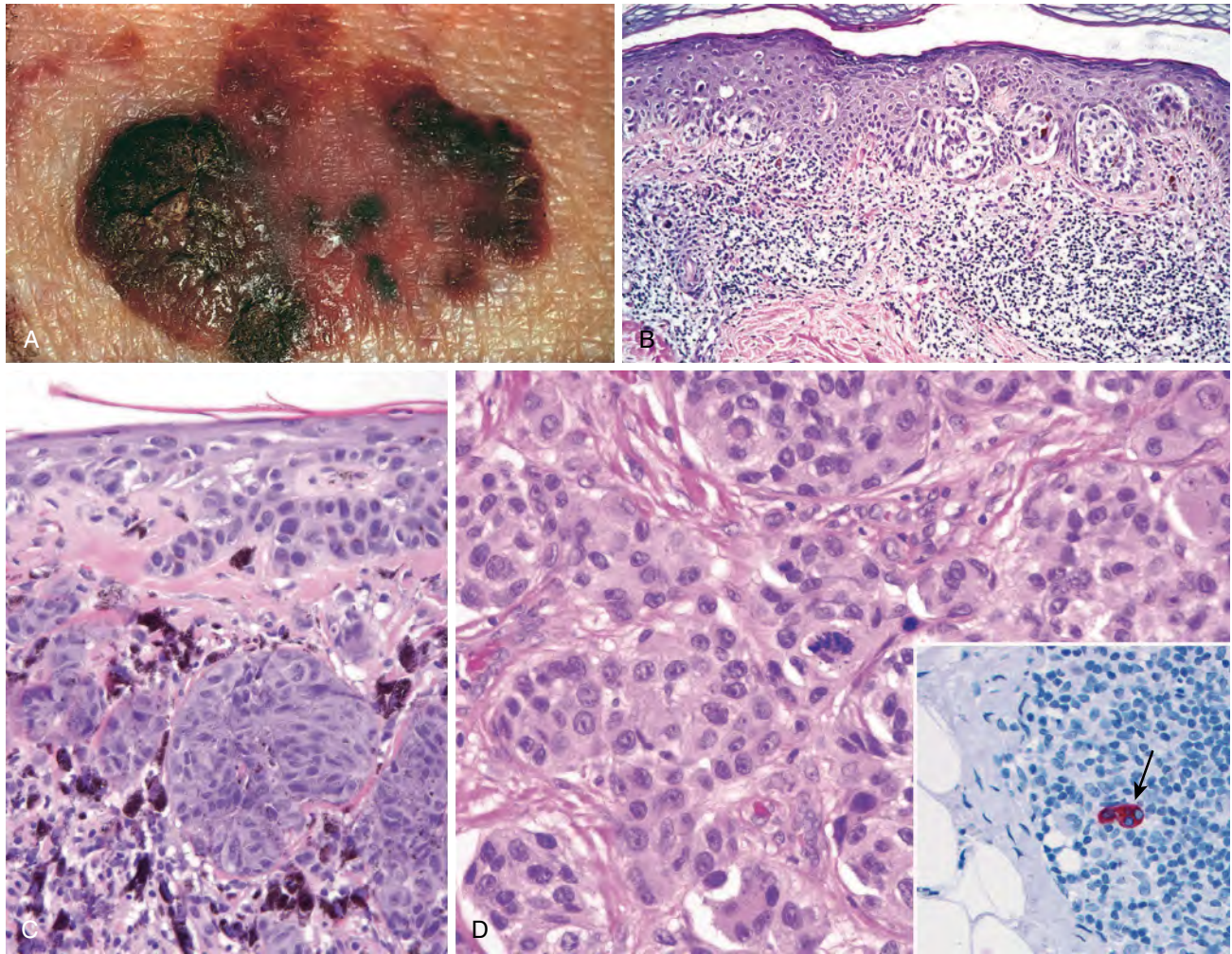


Fig. 25.8 Melanoma. (A) Typical lesions are irregular in contour and pigmentation. Macular areas correlate with the radial growth phase, while raised areas correspond to nodular aggregates of malignant cells in vertical growth phase. (B) Radial growth phase showing irregular nested and single-cell growth of melanoma cells within the epidermis and an underlying inflammatory response within the dermis. (C) Vertical growth phase demonstrating nodular aggregates of invasive tumor cells. (D) High-power view of melanoma cells. The *inset* shows a sentinel lymph node with a tiny cluster of melanoma cells (*arrow*) staining for the melanocytic marker HMB-45. Even small numbers of malignant cells in a draining lymph node may confer a worse prognosis.

on normal host tissues. Acral melanomas are more likely to have activating mutations in *KIT* than in *BRAF*, and a subset of *KIT*-mutated tumors responds to treatment with *KIT* tyrosine kinase inhibitors.

KEY CONCEPTS

Melanocytic Lesions

- Most melanocytic nevi have activating mutations in *BRAF*, or less often *RAS*, but the vast majority never undergo malignant transformation.
- Dysplastic nevi are best regarded as markers of melanoma risk rather than premalignant lesions.
- Melanoma is a highly aggressive malignancy linked to sun exposure; risk of spread is predicted by several tumor characteristics, particularly the vertical thickness of excised tumors.
- Activating *BRAF* mutations are common in cutaneous melanoma and predict response to inhibitors targeting *BRAF* and downstream kinases.
- Melanoma often incites a host immune response and can show dramatic responses to immune checkpoint therapies that enhance T-cell immunity.

BENIGN EPITHELIAL TUMORS

Benign cutaneous epithelial neoplasms are common tumors derived from the keratinizing stratified squamous epithelium of the epidermis and hair follicles and the ductular epithelium of cutaneous glands. They often recapitulate the structures from which they arise. Their appearance sometimes raises a concern of malignancy, particularly when they are pigmented or inflamed, and biopsy is frequently required to establish a definitive diagnosis. In rare instances they are a telltale sign of syndromes associated with potentially life-threatening visceral malignancies, such as multiple trichilemmomas in *Cowden syndrome* or multiple sebaceous neoplasms in *Muir-Torre syndrome*. Diagnosis of epithelial tumors in these instances can facilitate recognition of the underlying syndrome and implementation of appropriate clinical interventions.

Seborrheic Keratosis

These common epidermal tumors occur most frequently in middle-aged or older individuals. They arise spontaneously and are particularly

numerous on the trunk, although the extremities, head, and neck may also be involved.

Pathogenesis

Activating mutations in fibroblast growth factor receptor-3 (FGFR3), a receptor tyrosine kinase, are found in many sporadic seborrheic keratoses and are thought to drive the growth of the tumor. Seborrheic keratoses may suddenly appear in large numbers as part of a paraneoplastic syndrome (*Leser-Trélat sign*), possibly due to stimulation of keratinocytes by transforming growth factor- α (TGF- α) produced by tumor cells, most commonly carcinomas of the gastrointestinal tract.

MORPHOLOGY

Seborrheic keratosis characteristically appears as round to oval, flat, waxy, tan to dark brown plaque with a velvety or granular surface that varies in size from millimeters to several centimeters (Fig. 25.9A). In darker skinned individuals, seborrheic keratoses often appear as multiple small dark brown to black lesions on the face termed *dermatosis papulosa nigra*, a condition that is present in up to 35% of African American adults (Fig. 25.9B). Inspection with a hand lens typically reveals small, round, pore-like ostia impacted with keratin, a feature helpful in differentiating these pigmented lesions from melanoma.

On histologic examination, these neoplasms are exophytic and sharply demarcated from the adjacent epidermis. They are composed of sheets of small cells that closely resemble basal cells of the normal epidermis (Fig. 25.9C). Variable melanin pigmentation is present within these basaloid cells, accounting for the brown coloration. Exuberant keratin production (hyperkeratosis) occurs at the surface, and small keratin-filled cysts (**horn cysts**) and invaginations of keratin into the main mass (**invagination cysts**) are characteristic features. If irritated and inflamed, seborrheic keratoses develop whirling foci of squamous differentiation resembling eddy currents in a stream.

Acanthosis Nigricans

Acanthosis nigricans can be an important cutaneous sign of several underlying benign and malignant conditions. It is a condition marked by thickened, hyperpigmented skin with a velvet-like texture that most commonly appears in the flexural areas (axillae, skin folds of the neck, groin, and anogenital regions). It is divided into two types based on the underlying condition.

- *In at least 80% of cases, acanthosis nigricans is associated with benign conditions and develops gradually, usually during childhood or*

puberty. It may occur (1) as an autosomal dominant trait with variable penetrance, (2) in association with obesity or endocrine abnormalities (particularly with pituitary or pineal tumors, or with diabetes), and (3) as part of several rare congenital syndromes. The most common associations are with obesity and diabetes.

- *In the remaining cases, acanthosis nigricans arises in association with cancer, most commonly gastrointestinal adenocarcinoma, usually in middle-aged and older individuals. In this setting, acanthosis nigricans is best viewed as a paraneoplastic phenomenon that is likely caused by growth factors released from tumors.*

Pathogenesis

The unifying feature in all types of acanthosis nigricans is a disturbance that leads to increased growth factor receptor signaling in the skin. The familial form is associated with germline activating mutations in the receptor tyrosine kinase FGFR3, the same receptor that is frequently mutated in seborrheic keratosis. Depending on the mutation, acanthosis may be an isolated finding or be seen together with skeletal deformities, including achondroplasia and thanatophoric dysplasia. Why in some cases FGFR3 mutation gives rise to seborrheic keratosis and in others acanthosis nigricans is not clear. In those with type 2 diabetes, hyperinsulinemia is believed to provoke increased stimulation of insulin-like growth factor 1 receptor (IGF1R), another receptor tyrosine kinase that activates the same signaling pathways as FGFR3. Factors responsible for paraneoplastic acanthosis nigricans are uncertain; some cases have been linked to high levels of TGF- α , which may result in excessive activation of epidermal growth factor receptor (EGFR), yet another receptor tyrosine kinase, in the skin.

MORPHOLOGY

All forms of acanthosis nigricans have similar histologic features. The epidermis and underlying enlarged dermal papillae undulate sharply to form numerous repeating peaks and valleys. Variable hyperplasia may be seen, along with hyperkeratosis and slight basal cell layer hyperpigmentation (but no melanocytic hyperplasia).

Fibroepithelial Polyp

The fibroepithelial polyp has many names (acrochordon, squamous papilloma, skin tag) and is one of the most common cutaneous lesions. It usually comes to attention in middle-aged and older individuals on

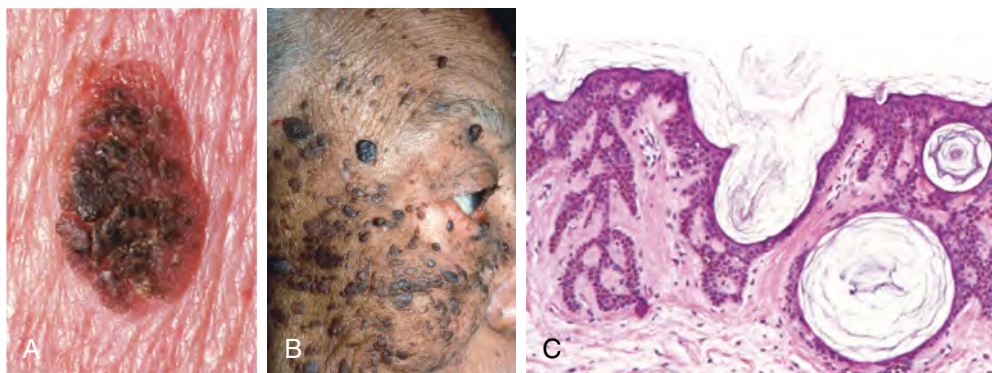


Fig. 25.9 Seborrheic keratosis. (A) A classic well-demarcated coin-like pigmented lesion with a “stuck on” appearance. (B) Dermatitis papulosa nigra is a variant of seborrheic keratosis seen in about one-third of African American adults. (C) Microscopically, the lesion is composed of benign basaloid cells associated with prominent keratin-filled “horn” cysts, some of which communicate with the surface (pseudohorn cysts). (B, From James WD, Elston DM, Treat JR, et al. *Andrews’ Diseases of the Skin*. 13th ed. Elsevier; 2020.)

the neck, trunk, face, and intertriginous areas. Rarely, fibroepithelial polyps and tumors of perifollicular mesenchyme (specialized fibroblasts associated with the hair bulb) are seen together in *Birt-Hogg-Dubé syndrome*, but most are sporadic.

MORPHOLOGY

Fibroepithelial polyps are soft, flesh-colored, bag-like tumors that are often attached to the surrounding skin by a slender stalk. On histologic examination, they consist of fibrovascular cores covered by benign squamous epithelium. It is not uncommon for the polyps to undergo ischemic necrosis due to torsion, which may cause pain and precipitate their removal.

Fibroepithelial polyps are usually inconsequential, but can occasionally be associated with diabetes, obesity, and intestinal polyposis. Of interest, like melanocytic nevi and hemangiomas, they often become more numerous or prominent during pregnancy, presumably related to hormonal stimulation.

Epithelial or Follicular Inclusion Cyst

Epithelial cysts are common lesions formed by the invagination and cystic expansion of the epidermis or, perhaps more commonly, a hair follicle. When large, they are susceptible to traumatic rupture, leading to spillage of keratin into the dermis that stimulates an extensive and often painful granulomatous inflammatory response.

ADNEXAL (APPENDAGE) TUMORS

There are literally hundreds of neoplasms arising from or showing differentiation toward cutaneous appendages. Their significance varies according to type and clinical context.

- Some are benign but mimic the appearance of cutaneous cancers such as basal cell carcinoma.
- Others are associated with germline mutations in tumor-suppressor genes. In such instances, the disorder sometimes presents with multiple adnexal tumors that may be disfiguring. In other instances, the associated appendage tumors are relatively trivial but warn of a predisposition for internal malignancy; such is the relationship between multiple *trichilemmomas* and *Cowden syndrome*, a disorder caused by germline mutations in the tumor-suppressor gene *PTEN* that is associated with an increased risk of endometrial cancer, breast cancer, and other malignancies.

Appendage tumors often appear as nondescript, flesh-colored solitary or multiple papules and nodules. Some have a predisposition to occur on specific body surfaces. Selected examples are provided here to illustrate neoplasms of hair follicles and sebaceous, eccrine, and apocrine glands.

- *Eccrine poroma* occurs predominantly on the palms and soles where sweat glands are numerous.
- *Cylindroma*, an appendage tumor with apocrine or eccrine differentiation, usually occurs on the forehead and scalp (Fig. 25.10A), where the coalescence of many nodules over time may produce a hat-like growth—hence the name *turban tumor*. Risk for cylindroma may be inherited as an autosomal dominant trait referred to *familial cylindromatosis*. In such cases, tumors appear early in life and are associated with germline inactivating mutations in the tumor-suppressor gene *CYLD*, which encodes a deubiquitinating enzyme that negatively regulates the oncogenic transcription factor NF- κ B. In some families, germline mutations in *CYLD* are associated with the occurrence of other appendage tumors as well, particularly trichoepithelioma and spiradenoma.

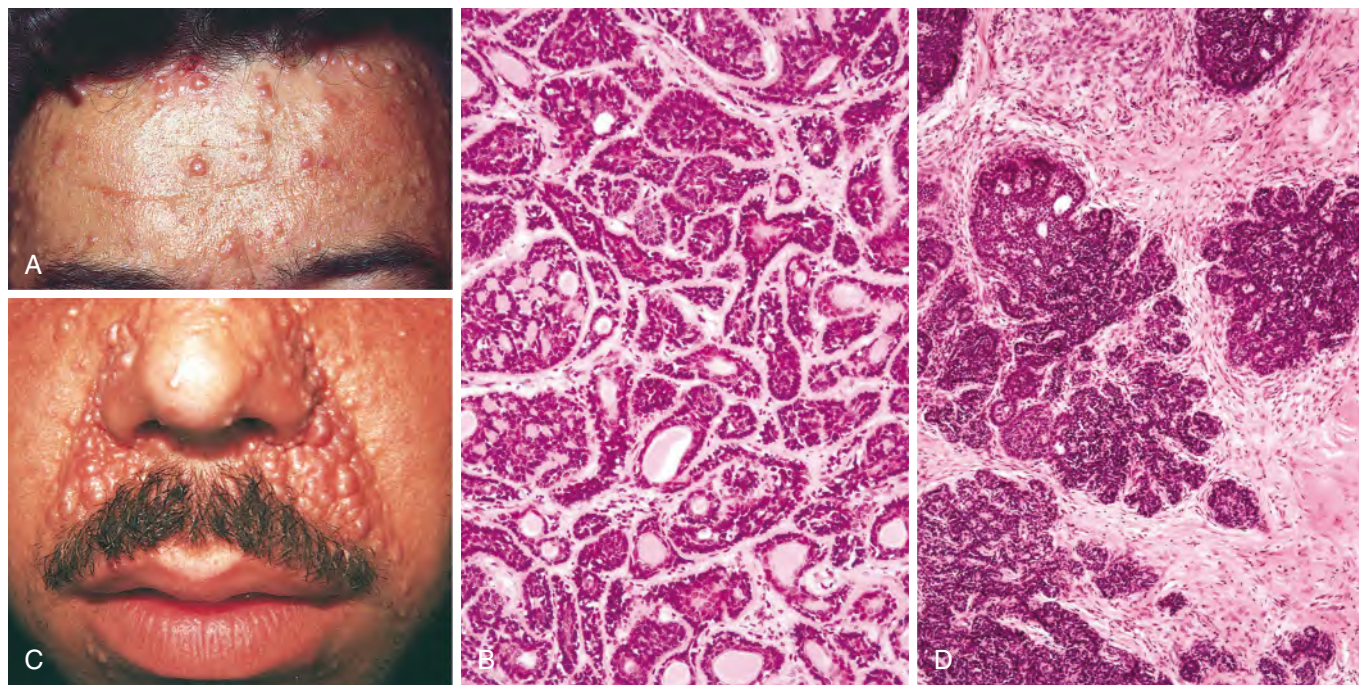


Fig. 25.10 Cylindroma and trichoepithelioma. (A, B) Multiple cylindromas (papules) on the forehead (A) are composed of islands of basaloid cells (B) containing occasional ducts that fit together like pieces of a jigsaw puzzle. (C–D) Perinasal papules and small nodules of trichoepithelioma (C) are composed of buds of basaloid cells (D) that resemble primitive hair follicles.

- *Syringoma*, a lesion with eccrine differentiation, usually occurs as multiple, small, tan papules in the vicinity of the lower eyelids. The name is derived from *syrinx*, the Greek word for pipe or tube.
- *Sebaceous adenoma* can be associated with internal malignancy in *Muir-Torre syndrome* and is also associated in a subset of cases with *Lynch syndrome* (Chapter 17), which is caused by germline defects in DNA mismatch repair genes.
- *Pilomatricoma*, a tumor with hair follicle differentiation, is associated with activating mutations in *CTNIB1*, the gene encoding β -catenin. Of note, Wnt signaling through β -catenin is critical for early hair development and regulates hair growth and maintenance.
- *Adnexal tumors can also show apocrine differentiation*; these usually arise in body areas where apocrine glands are most prevalent such as the axilla and scalp.

MORPHOLOGY

Cylindroma is composed of islands of cells resembling the normal epidermal or adnexal basal cell layer (basaloid cells). These islands fit together like pieces of a jigsaw puzzle within a fibrous dermal matrix (Fig. 25.10B). **Trichoepithelioma** is a proliferation of basaloid cells that forms primitive structures resembling hair follicles (Fig. 25.10D). **Sebaceous adenoma** is a lobular proliferation of sebocytes with increased peripheral basaloid cells and more mature sebocytes in the central portion that have frothy or bubbly cytoplasm due to the presence of lipid vesicles (Fig. 25.11A). **Pilomatricomas** are composed of basaloid cells that show trichilemmal or hair-like differentiation resembling that seen in the germinal portion of the normal hair bulb in the anagen growth phase (Fig. 25.11B). **Apocrine carcinoma** shows ductal differentiation with prominent decapitation secretion, similar to that seen in the normal apocrine gland (Fig. 25.11C).

Although most appendage tumors are benign, malignant variants exist. Apocrine tumors are unusual in that malignant forms appear to be more common than benign forms. *Sebaceous carcinoma* arises from the meibomian glands of the eyelid and may follow an aggressive course replete with systemic metastases. *Eccrine* and *apocrine carcinomas* can be confused with metastatic adenocarcinoma because of their tendency to form gland-like structures.

PREMALIGNANT AND MALIGNANT EPIDERMAL TUMORS

Actinic Keratosis

Actinic keratosis (as the name implies) usually occurs in sun-damaged skin and exhibits hyperkeratosis. As expected, these tumors have a particularly high incidence in lightly pigmented individuals. Exposure to ionizing radiation, industrial hydrocarbons, and arsenicals may induce similar lesions. These lesions may show progressively worsening dysplastic changes that culminate in cutaneous squamous cell carcinoma.

MORPHOLOGY

Actinic keratoses are usually less than 1 cm in diameter. They are typically tan-brown, red, or skin-colored and have a rough, sandpaper-like consistency. Some lesions produce so much keratin that a “cutaneous horn” develops (Fig. 25.12A), which in extreme cases may become so prominent that they resemble the actual horns of animals! Sun-exposed sites (face, arms, dorsum of hands) are most frequently affected. The lips may also develop similar lesions (termed **actinic cheilitis**).

Cytologic atypia is seen in the lowermost layers of the epidermis and may be associated with hyperplasia of basal cells (Fig. 25.12B) or, alternatively, with

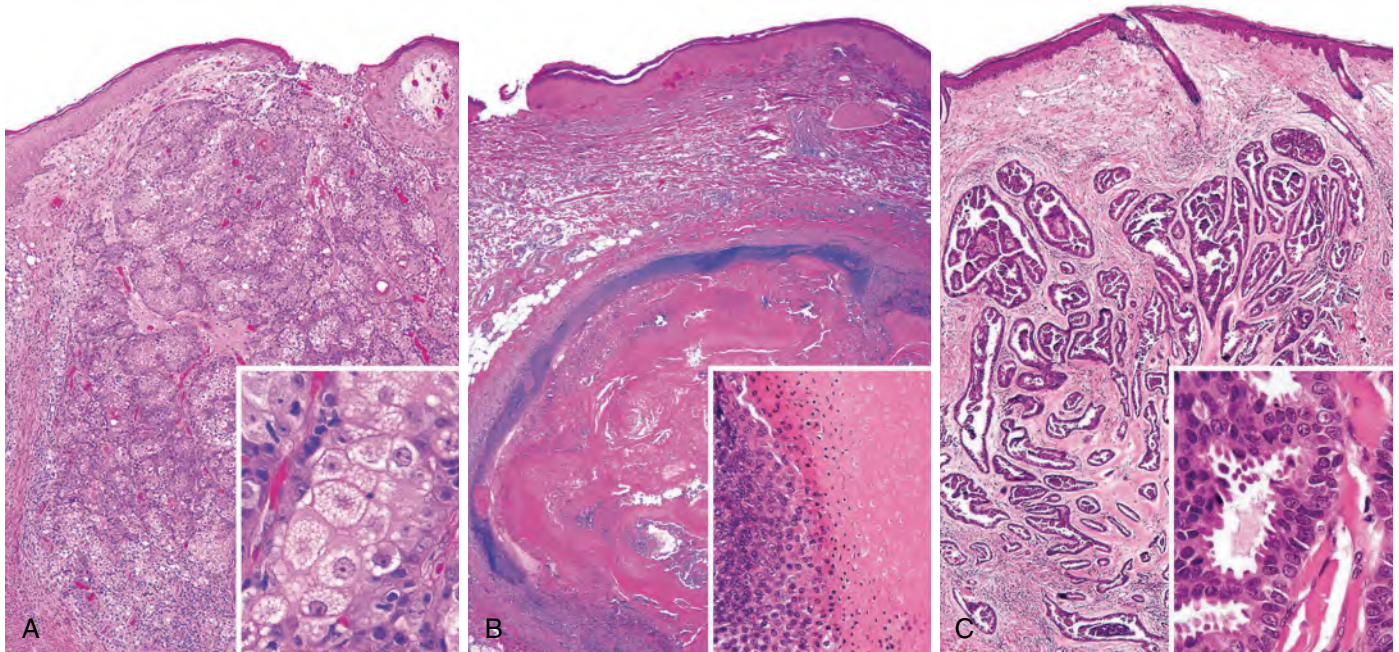


Fig. 25.11 Diverse adnexal tumors. (A) Sebaceous adenoma; *inset* demonstrates sebaceous differentiation. (B) Pilomatricoma; *inset* shows hair matrix differentiation to anucleate “ghost cells” (C) Apocrine carcinoma (well differentiated); *inset* shows apocrine differentiation and luminal secretions produced by decapitation of the lining cells.

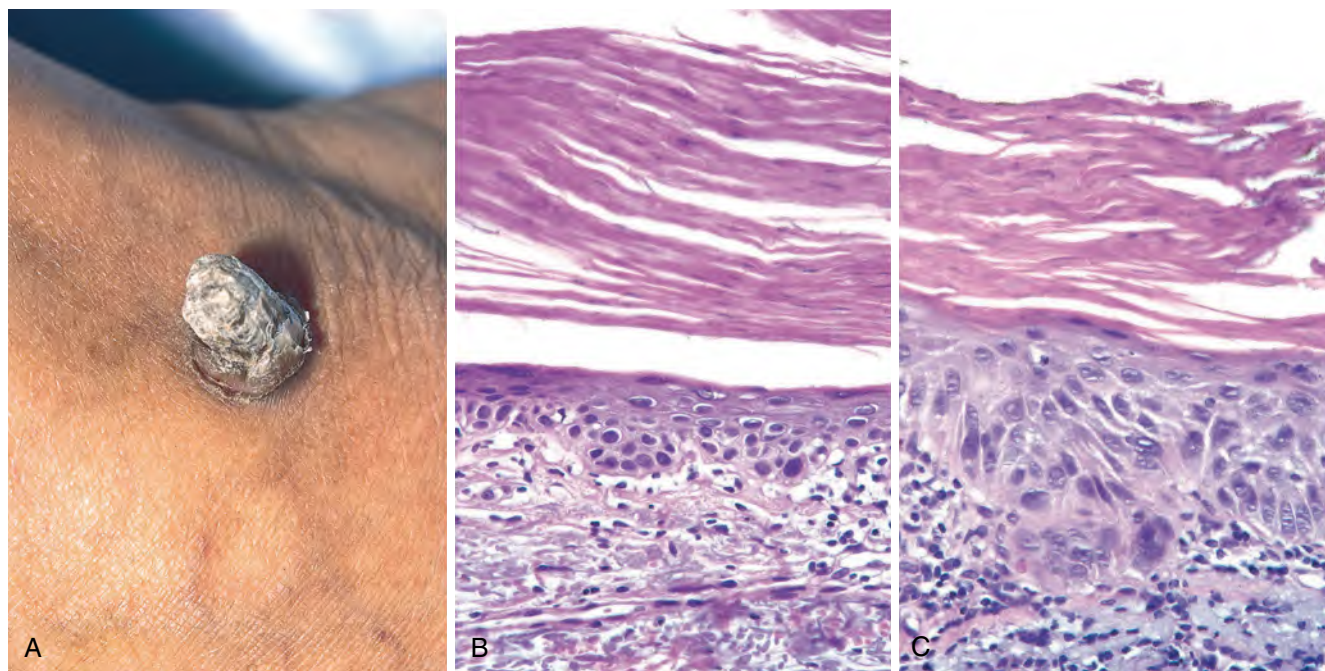


Fig. 25.12 Actinic keratosis. (A) Excessive keratotic scale in this lesion has produced a cutaneous horn. (B) Basal cell layer atypia (dysplasia) is associated with marked hyperkeratosis and parakeratosis. (C) Progression to full-thickness nuclear atypia, with or without the presence of superficial epidermal maturation, heralds the development of squamous cell carcinoma in situ.

atrophy that results in thinning of the epidermis. The atypical basal cells usually have pink or reddish cytoplasm due to dyskeratosis. Intercellular bridges are present, in contrast to basal cell carcinoma, in which they are not visible. The superficial dermis contains thickened, blue-gray elastic fibers (**elastosis**), the probable result of abnormal elastic fiber synthesis by sun-damaged fibroblasts. The stratum corneum is thickened and unlike normal skin, the cells in this layer often retain their nuclei (**parakeratosis**).

Actinic keratoses may regress or remain stable throughout life, but enough transform to malignancy that local eradication is warranted. This can be accomplished by gentle curettage, freezing, or topical application of chemotherapeutic agents. Of interest, topical administration of imiquimod, a drug that activates *toll-like receptors* (TLRs), eradicates up to half of lesions, presumably because activation of innate immunity triggers a host immune response to these precancerous lesions.

Squamous Cell Carcinoma

Squamous cell carcinoma is the second most common tumor arising on sun-exposed sites in older people, exceeded only by basal cell carcinoma. Except for lesions on the lower legs, these tumors have a higher incidence in males than in females. Invasive squamous cell carcinomas are usually discovered while they are small and resectable. Because they are usually easy to recognize and are treated effectively with local excision, fewer than 5% of these tumors metastasize to regional nodes, a frequency much lower than squamous cell carcinoma of other primary sites such as the oral mucosa and the lung. However, because squamous cell carcinoma of the skin is very common, metastatic disease takes a significant toll; it is estimated to cause as many as 8000 deaths per year, roughly equivalent to the number of deaths caused by metastatic melanoma.

Pathogenesis

The most important cause of cutaneous squamous cell carcinoma is DNA damage induced by exposure to UV light. Tumor incidence is proportional to the degree of lifetime sun exposure. A second common association is with immunosuppression, most notably in the setting of chemotherapy or organ transplantation. Immunosuppression may contribute to carcinogenesis by reducing host surveillance and possibly increasing the susceptibility of keratinocytes to infection and transformation by viruses. Among those suspected to be involved are human papillomavirus (HPV) subtypes 5 and 8, but these viruses have only been clearly implicated in tumors arising in patients with the rare autosomal recessive condition *epidermodysplasia verruciformis*, which is marked by a high susceptibility to cutaneous squamous cell carcinomas. In addition to its damaging effect on DNA, sunlight, through uncertain mechanisms, seems to cause a transient defect in cutaneous innate immunity that may diminish immune-mediated elimination of sun-damaged cells. Other risk factors for squamous cell carcinoma include industrial carcinogens (tars and oils), chronic ulcers and draining osteomyelitis, old burn scars, ingestion of arsenicals, ionizing radiation, and (in the oral cavity) tobacco and betel nut chewing.

Most studies on the genetics of squamous cell carcinoma have focused on acquired defects in sporadic tumors and their precursors (actinic keratoses) and the relationships between these defects and sun exposure. The incidence of *TP53* mutations in actinic keratoses found in lightly pigmented individuals is high, suggesting that p53 dysfunction is an early event in the development of tumors induced by sunlight. Of note, the mutations that are seen in *TP53* often occur at pyrimidine dimers, indicating that they, too, stem from damage caused by UV light. A similar story underlies the remarkable susceptibility of patients with *xeroderma pigmentosum* to squamous cell carcinoma. This disorder is caused by inherited mutations in genes in the nucleotide excision repair pathway, which is required for accurate

repair of pyrimidine dimers; when this pathway is defective, error-prone repair pathways take over, leading to the rapid accumulation of mutations and carcinogenesis.

As with all forms of cancer, cutaneous squamous cell carcinoma stems from multiple driver mutations. In addition to defects in p53, mutations that increase RAS signaling and decrease Notch signaling are common contributors to the transformation process.

MORPHOLOGY

Squamous cell carcinomas that have not invaded through the basement membrane of the dermoepidermal junction (in situ carcinoma) appear as sharply defined, red, scaling plaques. More advanced, invasive lesions are nodular, show variable keratin production (appreciated grossly as hyperkeratotic scale), and may ulcerate (Fig. 25.13A).

Unlike actinic keratoses, in squamous cell carcinoma, in situ cells with atypical (enlarged and hyperchromatic) nuclei involve all levels of the epidermis (Fig. 25.12C). Invasive squamous cell carcinoma (Fig. 25.13B,C) shows variable degrees of differentiation, ranging from tumors composed of polygonal cells arranged in orderly lobules and having numerous large areas of keratinization to neoplasms consisting of highly anaplastic cells that exhibit only abortive, single-cell keratinization (dyskeratosis). The latter tumors may be so poorly differentiated that immunohistochemical stains for keratins are needed to confirm the diagnosis.

Metastatic cutaneous squamous cell carcinoma is resistant to chemotherapy and sensitive to radiotherapy, which is used to treat localized metastatic disease. Like melanoma, cutaneous squamous cell carcinoma has a very high mutational burden and responds well to immune checkpoint inhibitors, which are now often chosen as the first line treatment for patients with metastatic disease.

Basal Cell Carcinoma

Basal cell carcinoma is a distinctive locally aggressive cutaneous tumor associated with mutations that activate the Hedgehog signaling pathway. Basal cell carcinoma is the most common invasive cancer in humans, numbering nearly 1 million cases per year in the United States. These are slow-growing tumors that rarely metastasize. The vast majority are recognized at an early stage and cured by local excision. However, a small fraction of tumors (<0.5%) are locally aggressive and potentially disfiguring, and on rare occasions metastasis to distant sites is seen. Basal cell carcinoma occurs at sun-exposed sites in lightly pigmented older adults. As with squamous cell carcinoma, the incidence of basal cell carcinoma is increased in the setting of immunosuppression and in disorders of DNA repair, such as *xeroderma pigmentosum* (Chapter 7).

Pathogenesis

As is often the case in biology and medicine, study of a rare genetic syndrome associated with a high risk of a common disease (basal cell carcinoma) led to the elucidation of pathogenic mechanisms of general importance. The syndrome in question, *nevroid basal cell carcinoma syndrome* (NBCCS) (also known as basal cell nevus syndrome or *Gorlin syndrome*), is an autosomal dominant disorder characterized by the development of multiple basal cell carcinomas, often before age 20 years, accompanied by various other tumors (especially medulloblastomas and ovarian fibromas), odontogenic keratocysts, pits of the palms and soles, and certain developmental abnormalities. NBCCS is one of several cancer syndromes associated with skin manifestations (Table 25.3). The gene associated with NBCCS is *PTCH*, a tumor-suppressor gene that is the human homologue of the *Drosophila* developmental gene *patched*. Individuals with NBCCS are born with a germline loss-of-function mutation in one *PTCH* allele; the second normal allele is inactivated in tumors

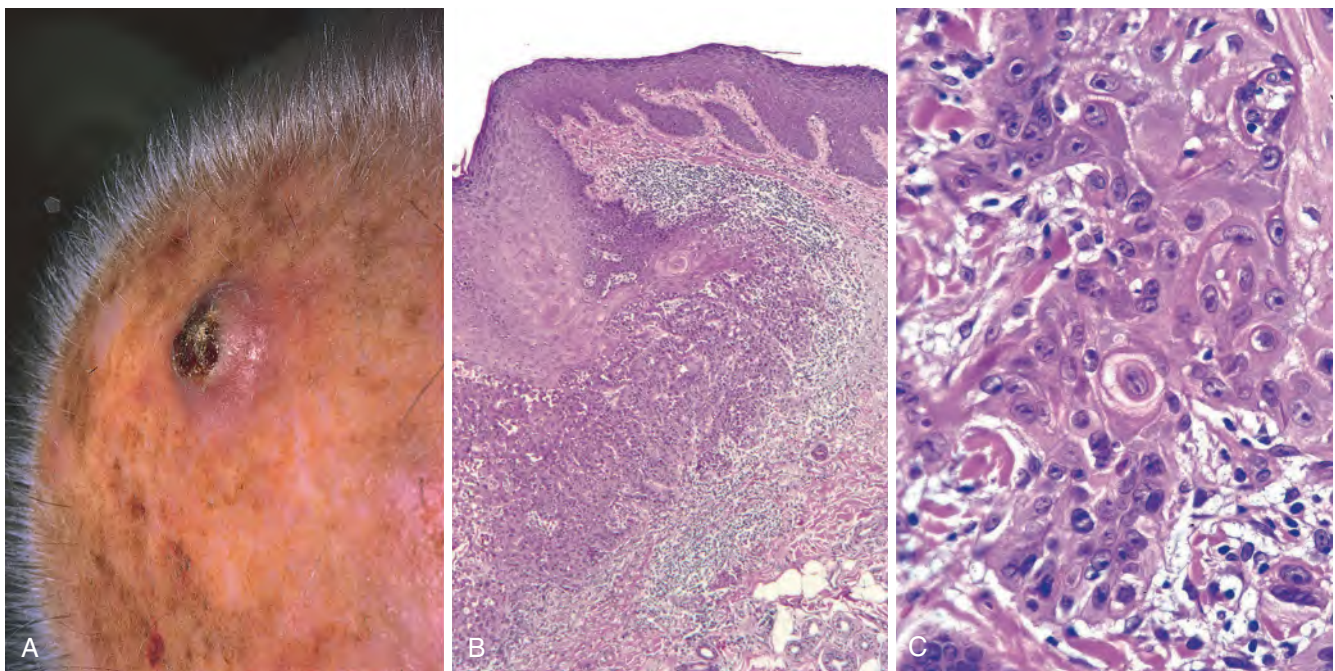


Fig. 25.13 Invasive squamous cell carcinoma. (A) Lesions are often nodular and ulcerated, as seen in this scalp tumor. (B) Tongues of atypical squamous epithelium have transgressed the basement membrane and invaded deeply into the dermis. (C) Invasive tumor cells show enlarged nuclei with angulated contours and prominent nucleoli.

TABLE 25.3 Survey of Familial Cancer Syndromes With Cutaneous Manifestations

Disease	Inheritance	Chromosomal Location	Gene/Protein	Normal Function/Manifestation of Loss
Ataxia-telangiectasia	AR	11q22.3	<i>ATM/ATM</i>	DNA repair after radiation injury/neurologic and vascular lesions
Nevoid basal cell carcinoma syndrome	AD	9q22	<i>PTCH/PTCH</i>	Developmental patterning gene/multiple basal cell carcinomas; medulloblastoma; jaw cysts
Cowden syndrome	AD	10q23	<i>PTEN/PTEN</i>	Lipid phosphatase/benign follicular appendage tumors (trichilemmomas); internal adenocarcinoma (often breast or endometrial)
Familial atypical multiple mole-melanoma syndrome	AD	9p21	<i>CDKN2A/p16</i> , <i>INK4</i> <i>CDKN2A/p14</i> , <i>ARF</i>	Inhibits CDK4/6 phosphorylation of RB, promoting cell cycle arrest/melanoma; pancreatic carcinoma Binds MDM2, promoting p53 function/melanoma; pancreatic carcinoma
Muir-Torre syndrome	AD	2p22 3p21	<i>MSH2/MSH2</i> <i>MLH1/MLH1</i>	Involved in DNA mismatch repair/sebaceous neoplasia; internal malignancy (colon and others)
Neurofibromatosis 1	AD	17q11	<i>NF1/neurofibromin</i>	Negatively regulates RAS signaling/neurofibromas
Neurofibromatosis 2	AD	22q12	<i>NF2/merlin</i>	Integrates cytoskeletal signaling/neurofibromas; acoustic neuromas
Tuberous sclerosis	AD	9q34 16p13	<i>TSC1/hamartin</i> <i>TSC2/tuberlin</i>	Work together in a complex that negatively regulates mTOR/angiofibromas; intellectual disability
Xeroderma pigmentosum	AR	9q22 and others	<i>XPA/XPA</i> and others	Nucleotide excision repair/melanoma; nonmelanoma skin cancers

AD, Autosomal dominant; AR, autosomal recessive.

From Tsai KY, Tsao H. The genetics of skin cancer. *Am J Med Genet C Semin Med Genet.* 2004;131C:82-92.

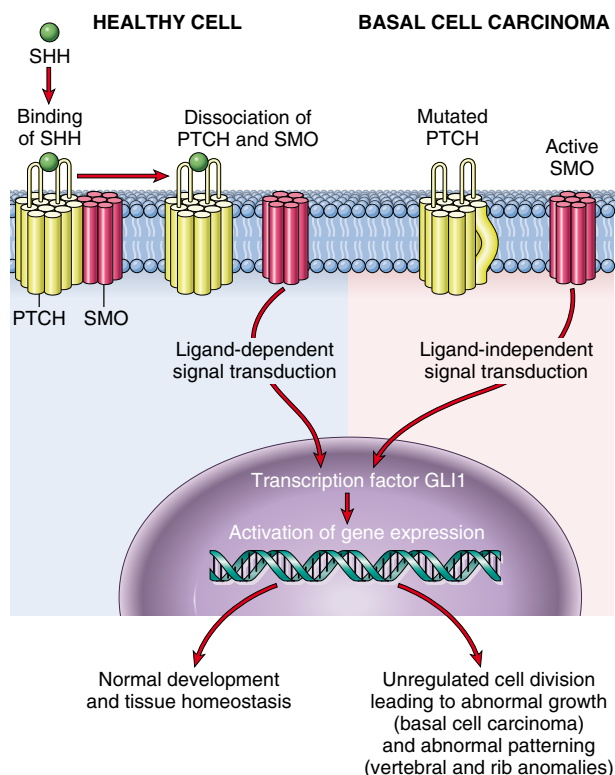


Fig. 25.14 Normal and oncogenic Hedgehog signaling. *Left*, Normally, PTCH and SMO form a receptor complex that binds sonic hedgehog (SHH). In the absence of SHH, PTCH blocks SMO activity. When SHH binds PTCH, SMO is released to trigger a signal transduction cascade that leads to activation of GLI1 and other transcription factors. *Right*, Mutations in *PTCH*, and less often in *SMO*, allow SMO to signal without SHH binding and produce constitutive activation of GLI1. GLI signaling is a characteristic feature of sporadic basal cell carcinomas and tumors associated with nevoid basal cell carcinoma (Gorlin) syndrome.

by an acquired mutation, usually caused by exposure to mutagens (particularly UV light).

PTCH protein is a receptor for *sonic hedgehog* (SHH), a component of the hedgehog signaling pathway, which controls polarity and central nervous system development during embryogenesis and regulates hair follicle formation and hair growth. In the “off” state, PTCH exists in a complex with another transmembrane protein called SMO (for “smoothed”). Binding of SHH to PTCH releases SMO, which in turn activates the transcription factor GLI1 (Fig. 25.14), thus turning on the expression of genes that support tumor cell growth and survival. Mice engineered to have excessive GLI1 activation are prone to develop skin tumors resembling basal cell carcinomas. Similarly, in NBCCS the loss of PTCH function causes constitutive activation of SMO and GLI1, leading to the development of basal cell carcinoma.

Mutations that activate hedgehog signaling are also prevalent in sporadic basal cell carcinomas. Loss-of-function *PTCH* mutations are common, and about one-third of these mutations consist of C→T transitions that are hallmarks of UV damage. This insight has paved the way for the development and clinical implementation of small molecule inhibitors of the Hedgehog pathway.

MORPHOLOGY

Basal cell carcinomas usually present as **pearly papules** containing prominent dilated subepidermal blood vessels (**telangiectasias**) (Fig. 25.15A), features that may be less prominent in individuals with more highly pigmented skin (Fig. 25.15B). Some tumors contain melanin and superficially resemble melanocytic nevi or melanomas. Advanced lesions may ulcerate, and extensive local invasion of bone or facial sinuses may occur after many years of neglect, explaining the archaic designation **rodent ulcers**. One common and important variant, the superficial basal cell carcinoma, presents as an erythematous, occasionally pigmented plaque that may mimic the appearance of melanoma.

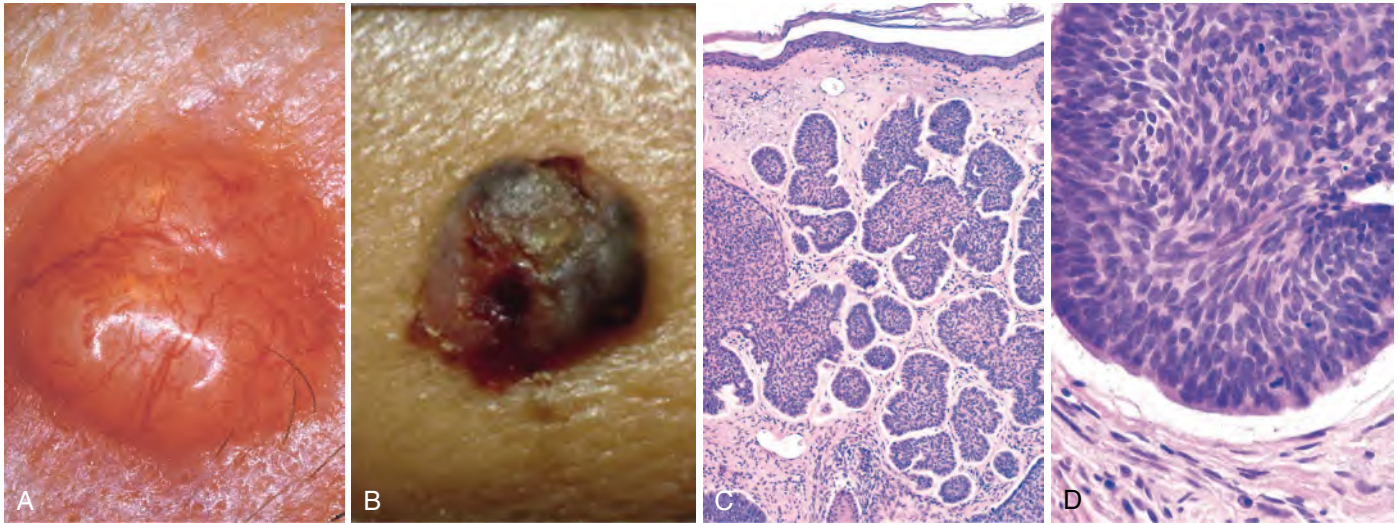


Fig. 25.15 Basal cell carcinoma. Pearly, telangiectatic nodules on lighter skin (A) are sometimes less prominent on darker skin (B). Both are composed of nests of uniform basaloid cells within the dermis (C) that are often separated from the adjacent stroma by thin clefts (D), an artifact of sectioning. (B, From DermNet NZ, 2024. www.dermnetnz.org.)

Histologically, the tumor cells resemble those in the normal basal cell layer of the epidermis. They arise from the epidermis or follicular epithelium and do not occur on mucosal surfaces. Two patterns are seen: **multifocal growths** originating from the epidermis and sometimes extending over several square centimeters or more of skin surface (multifocal superficial type) and **nodular lesions** growing downward deeply into the dermis as cords and islands of variably basophilic cells with hyperchromatic nuclei, embedded in a mucinous matrix and often surrounded by many fibroblasts and lymphocytes (Fig. 25.15C). The cells at the periphery of the tumor cell islands tend to be arranged radially with their long axes in parallel alignment (**palisading**). In sections, the stroma retracts away from the carcinoma (Fig. 25.15D), creating artifactual clefts that assist in differentiating basal cell carcinomas from certain appendage tumors that are also characterized by proliferation of basaloid cells, such as trichoepithelioma.

Most basal cell carcinomas are cured by local excision. Rare cases of disease that is locally advanced or metastatic often respond well to Hedgehog pathway inhibitors or treatment with immune checkpoint inhibitors.

Merkel Cell Carcinoma

Merkel cell carcinoma, a rare, aggressive skin cancer, merits brief mention because of its association with Merkel cell virus (MCV), a member of the polyomavirus family. The cell of origin for these cancers is thought by most investigators to be the Merkel cell, a specialized type of epithelial cell involved in light touch sensation that may also have neuroendocrine functions. Merkel cell carcinoma occurs in two distinct clinical settings: (1) patients who are chronically immunosuppressed and (2) older, lightly pigmented patients with a history of sun exposure. As you might surmise, tumors associated with MCV infection mainly occur in immunosuppressed patients, such as individuals with AIDS or who have undergone organ transplantation.

Like other transforming DNA viruses, MCV encodes viral proteins that interfere with or alter the function of key host cell tumor suppressors and oncoproteins, specifically Rb, p53, PP2A (a protein phosphatase with tumor suppressive activity), and MYC. Serologic studies have shown that by adulthood virtually all individuals have been infected by MCV, yet Merkel cell carcinoma is rare, presumably because T cells specific for MCV proteins protect against development of MCV-associated Merkel cell carcinoma. Sequencing of the genomes of MCV-associated

Merkel cell carcinoma has shown that the viral DNA is integrated into the cancer cell genome, which carries only a small number of acquired mutations and is usually diploid. By contrast, cancers arising in immunocompetent individuals carry a heavy burden of mutations bearing the signature of UV light-mediated DNA damage and commonly have acquired loss of function mutations in *RB* and *TP53* and numerous copy number gains and losses.

In both clinical contexts, Merkel cell carcinoma typically presents as a rapidly enlarging cutaneous nodule. Microscopically, the tumor resembles an aggressive neuroendocrine carcinoma (Chapter 24). Merkel cell carcinoma is highly responsive to radiation therapy and chemotherapy, but spread to regional lymph nodes is common and chemoradiation therapy is rarely curative. Notably, tumors arising in sun-damaged skin are highly responsive to immune checkpoint inhibitors, as is typical of cancers with a high burden of somatic mutations, and some patients treated with checkpoint inhibitors, including those with distant metastatic disease, have sustained responses and may even be cured.

KEY CONCEPTS

Malignant Epidermal Tumors

- The incidence of both squamous cell carcinoma and basal cell carcinoma is strongly correlated with increasing lifetime sun exposure.
- Cutaneous squamous cell carcinoma can progress from actinic keratoses but also arises from chemical exposures, at thermal burn sites, and in the setting of immunosuppression.
- Cutaneous squamous cell carcinoma has potential for metastasis but does so much less frequently than squamous cell carcinoma arising at other sites (e.g., oral mucosa, lung).
- Basal cell carcinoma, the most common malignancy worldwide, is a locally aggressive tumor associated with mutations that activate Hedgehog signaling. Metastasis is very rare.
- Merkel cell carcinoma is a rare, aggressive skin tumor that arises through two different pathways: (1) Merkel cell virus infection, in the context of immunosuppression; and (2) DNA damage caused by exposure to sunlight.

TUMORS OF THE DERMIS

The dermis contains a variety of elements such as smooth muscle, pericytes, fibroblasts, neural tissue, and endothelium. Neoplasms composed

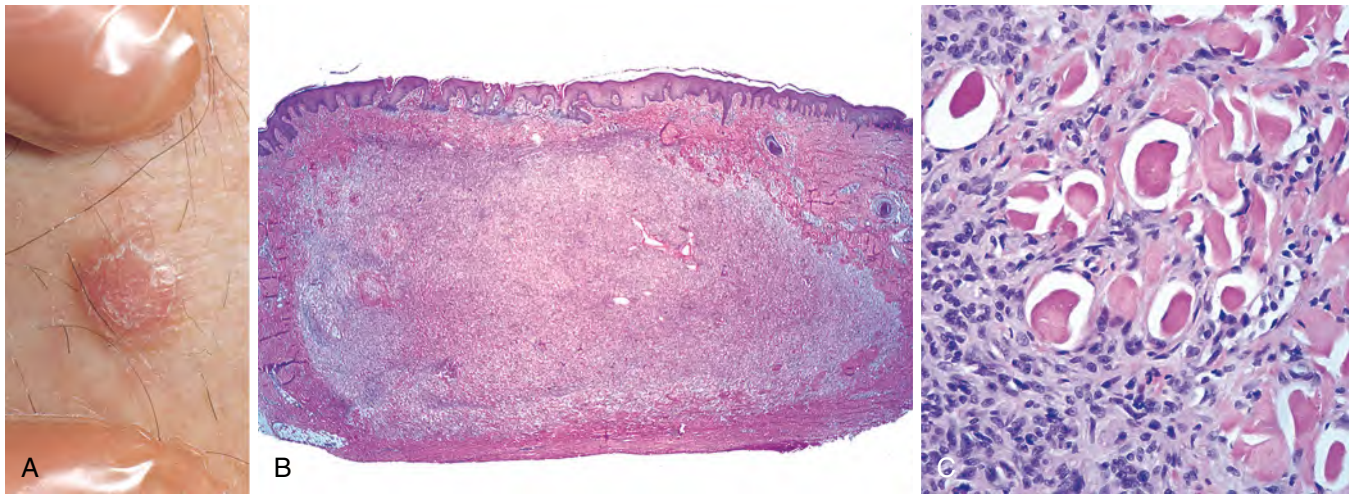


Fig. 25.16 Benign fibrous histiocytoma (dermatofibroma). This firm, tan papule on the leg (A) contains a circumscribed dermal proliferation of benign-appearing spindle cells (B). Note the characteristic overlying epidermal hyperplasia (B) and the tendency of fibroblasts to surround individual collagen bundles (C).

of cells resembling all these elements occur in the skin, but most also involve other soft tissues and viscera and are discussed elsewhere or are too rare to merit mention. This section discusses two dermal neoplasms—one benign, one malignant—that arise primarily in the skin.

Benign Fibrous Histiocytoma (Dermatofibroma)

Benign fibrous histiocytoma refers to a family of morphologically and histogenetically related benign dermal neoplasms of uncertain lineage. These tumors are usually seen in adults and often occur on the legs of young and middle-aged females. Lesions are asymptomatic or tender and may increase and decrease slightly in size over time. Their biologic behavior is indolent.

The cause of fibrous histiocytomas remains a mystery. Some cases have a history of antecedent trauma, suggesting an abnormal response to injury and inflammation, perhaps analogous to the deposition of increased amounts of altered collagen in a hypertrophic scar or keloid. However, several fusion genes encoding portions of protein kinases such as the receptor tyrosine kinase ALK or protein kinase C (PKC) have been identified in a subset of cases, suggesting that these proliferations are best considered true neoplasms. These tumors appear to be composed of at least partially of dermal dendritic cells.

MORPHOLOGY

These neoplasms appear as firm, tan to brown papules (Fig. 25.16A). Most are less than 1 cm in diameter, but actively growing lesions may reach several centimeters in diameter; with time they often become flattened.

The most common form of fibrous histiocytoma is referred to as a **dermatofibroma**. These tumors consist of benign, spindle-shaped cells that are usually arranged in a well-defined, nonencapsulated mass within the mid-dermis (Fig. 25.16B,C). Extension of these cells into the subcutaneous fat is sometimes observed. Many cases demonstrate a peculiar form of overlying epidermal hyperplasia, characterized by downward elongation of hyperpigmented rete ridges (**pseudoepitheliomatous hyperplasia**). Numerous histologic variants are noted, such as more cellular forms or tumors with pools of extravascular blood and hemosiderin (aneurysmal variants).

Dermatofibrosarcoma Protuberans

Dermatofibrosarcoma protuberans is best regarded as a well-differentiated fibrosarcoma of the skin. These tumors are slow growing.

They may be locally aggressive and can recur after excision, but they rarely metastasize. Metastasis is most often seen with tumors that exhibit greater cytologic atypia.

Pathogenesis

The molecular hallmark of dermatofibrosarcoma protuberans is a **translocation involving the genes encoding collagen1A1 (COL1A1) and platelet-derived growth factor-β (PDGFB)**. The resulting rearrangement juxtaposes the COL1A1 promoter and the coding region of PDGFB and leads to overexpression and increased secretion of PDGF-β, which drives tumor cell growth through an autocrine loop. While the primary mode of treatment is wide local excision, rare cases that are unresectable due to their location or because of metastatic spread can be treated with inhibitors of the PDGF-β receptor tyrosine kinase. Withdrawal of tyrosine kinase inhibitor therapy leads to regrowth of the tumor, so treatment must be lifelong.

MORPHOLOGY

Dermatofibrosarcoma protuberans usually appears as a “protuberant” nodule, most often on the trunk, within a firm (indurated) plaque that may sometimes ulcerate (Fig. 25.17A). In contrast to dermatofibroma, the overlying epidermis is generally thinned. Deep extension from the dermis into subcutaneous fat, producing a characteristic “Swiss cheese” pattern, is frequently seen (Fig. 25.17B). The tumors are composed of closely packed fibroblasts arranged radially so as to resemble the blades of a pinwheel, a pattern referred to as **storiform** (Fig. 25.17C). Mitoses are rare.

TUMORS OF CELLULAR MIGRANTS TO THE SKIN

Aside from tumors that arise directly from epidermal and dermal cells, several proliferative disorders of the skin involve cells whose progenitors arise elsewhere and then specifically home to the cutaneous microenvironment.

Mycosis Fungoides (Cutaneous T-Cell Lymphoma)

Cutaneous T-cell lymphoma (CTCL) spans a spectrum of lymphoproliferative disorders affecting the skin (Chapter 13), many with distinctive presentations. This section focuses on *mycosis fungoides*, a lymphoma of skin-homing CD4+ helper T cells that presents in the

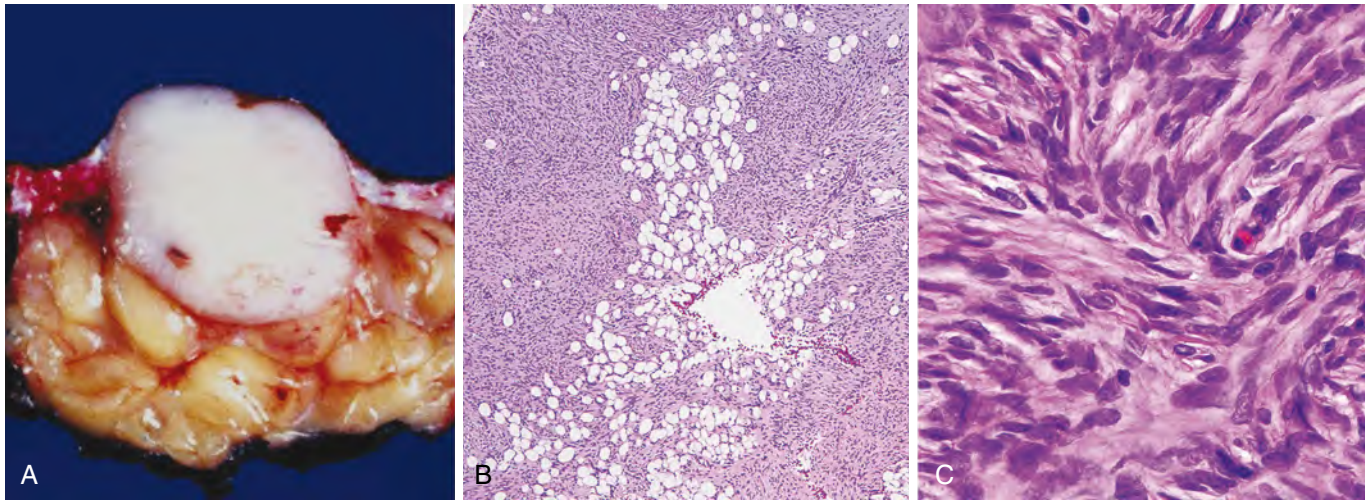


Fig. 25.17 Dermatofibrosarcoma protuberans. (A) The tumor consists of a flesh-colored fibrotic nodule on sectioning. (B) The lesion often infiltrates the subcutis in a manner reminiscent of “Swiss cheese” to aficionados. (C) Characteristic storiform (swirling) alignment of the spindle cells is apparent.

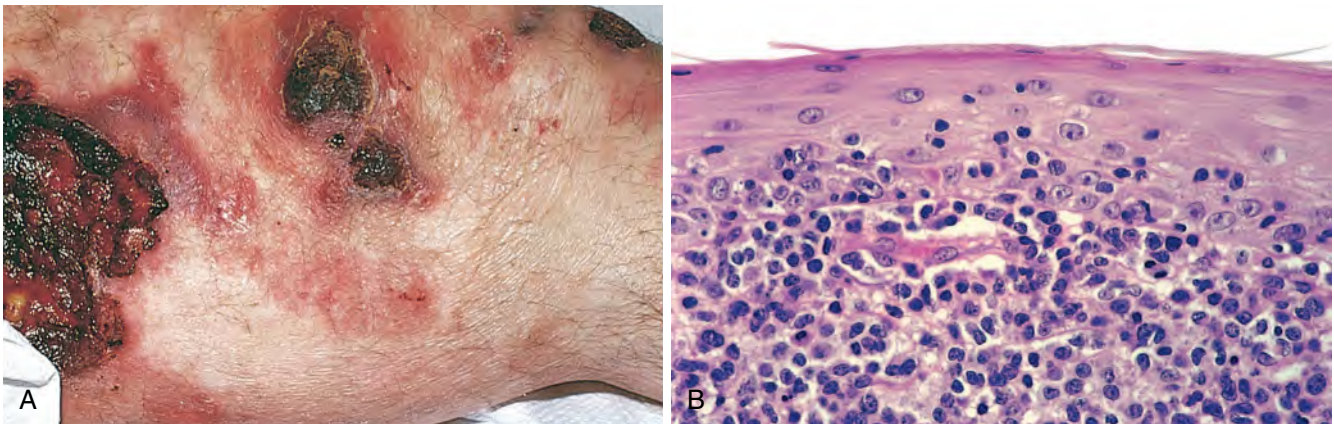


Fig. 25.18 Cutaneous T-cell lymphoma (mycosis fungoides). (A) Several erythematous plaques with scaling and ulceration are evident. (B) Microscopically, there is an infiltrate of atypical lymphocytes that accumulates beneath and invades the epidermis.

skin. In most affected individuals, the disease remains localized to the skin for many years, but it may eventually evolve into a systemic lymphoma. This tumor may occur at any age, but most commonly afflicts persons older than age 40 years.

Lesions of mycosis fungoides usually involve truncal areas and have diverse appearances, including scaly, red-brown patches; raised, scaling plaques that may be confused with psoriasis; and fungating nodules. Prognosis is related to the percentage of body surface involved and the presence of tumor nodules or systemic spread, both of which are associated with aggressive disease. Eczema-like lesions typify early stages of disease when obvious visceral or nodal spread has not occurred. Raised, indurated, irregularly outlined erythematous plaques may then supervene. Development of multiple tumor nodules correlates with systemic spread. Sometimes the plaques and nodules ulcerate (Fig. 25.18A). Ultimately, lesions may affect numerous body surfaces including the trunk, extremities, face, and scalp.

The proliferating cells in CTCL are usually clonal populations of CD4+ helper T cells that home to the skin due to expression of cutaneous lymphocyte antigen. A wide variety of driver gene mutations in

oncogenes and tumor-suppressor genes have been described, none of which are specific for CTCL. The neoplastic cells have clonal T-cell receptor gene rearrangements and sometimes express aberrant combinations of T-cell surface antigens. Topical therapy with steroids or UV light is often used for early skin lesions, whereas more aggressive systemic chemotherapy is indicated for advanced disease.

MORPHOLOGY

The histologic hallmark of mycosis fungoides is the presence of atypical cells that characteristically form band-like aggregates within the superficial dermis (Fig. 25.18B) and invade the epidermis as single cells and small clusters (**Pautrier microabscesses**). These cells often have markedly infolded nuclear membranes, imparting a convoluted or cerebriform contour. Although patches and plaques show pronounced epidermal infiltration by neoplastic cells (epidermotropism), in more advanced nodular lesions the malignant T cells often lose their epidermotropism, grow deeply into the dermis to produce nodules, and eventually spread systemically.

Mastocytosis

The term *mastocytosis* encompasses a spectrum of rare disorders characterized by increased numbers of mast cells in the skin and, in some instances, in other organs as well. A cutaneous form of the disease that affects predominantly children and accounts for more than 50% of all cases is termed *urticaria pigmentosa*. The cutaneous lesions are usually multiple, although solitary mastocytomas may also occur in very young children. About 10% of individuals with mast cell disease have systemic disease, with mast cell infiltration of many organs. These individuals are often adults, and unlike those with localized cutaneous disease, the prognosis in individuals with systemic disease is more guarded.

Many of the signs and symptoms of mastocytosis are due to the release of histamine, heparin, and other substances when mast cells degranulate. *Darier sign* refers to a localized area of dermal edema and erythema (wheal) that occurs when lesional skin is rubbed. *Dermatographism* refers to an area of dermal edema resembling a hive that occurs following localized stroking of apparently normal skin with a pointed instrument. In systemic disease, all of the following may be seen: pruritus and flushing, triggered by certain foods, temperature changes, alcohol, and certain drugs (morphine, codeine, aspirin); watery nasal discharge (rhinorrhea); rarely, gastrointestinal or nasal bleeding, possibly due to the anticoagulant effects of heparin; and bone pain, which may be caused by mast cell infiltration or by pathologic fractures stemming from osteoporosis. Osteoporosis is caused by excessive histamine release in the marrow microenvironment and can be a clue to the diagnosis, particularly in premenopausal females and in males. A clue to the diagnosis is an elevated serum level of mast cell tryptase, which is released into the blood from the neoplastic mast cells.

Pathogenesis

Many cases of mastocytosis have acquired activating point mutations in the KIT receptor tyrosine kinase. The resulting increase in KIT signaling drives mast cell growth and survival. This insight has

led to the clinical development of KIT kinase inhibitors, which often produce dramatic tumor regression, even in patients with advanced aggressive systemic disease.

MORPHOLOGY

The pathologic findings are highly variable. In *urticaria pigmentosa*, lesions are multiple and widely distributed, consisting of round to oval, red-brown, nonscaling papules and small plaques. Solitary *mastocytoma* presents as a pink to tan-brown nodule that may be pruritic or show blister formation (Fig. 25.19A). The histologic picture in urticaria pigmentosa or solitary mastocytoma varies from a subtle increase in mast cells around superficial dermal blood vessels to large numbers of tightly packed mast cells in the upper to mid-dermis (Fig. 25.19B). Fibrosis, edema, and eosinophils may also be present. Mast cells may be difficult to differentiate from lymphocytes in routine, hematoxylin and eosin–stained sections, and special metachromatic stains (toluidine blue or Giemsa) are used to visualize their granules (Fig. 25.19C). Even with these stains, extensive degranulation may result in failure to recognize these cells by light microscopy, but their identity can be readily confirmed with immunohistochemical stains for mast cell markers, such as mast cell tryptase and KIT.

DISORDERS OF EPIDERMAL MATURATION

Ichthyosis

Of the numerous disorders that impair epidermal maturation, ichthyosis is one of the most striking. The term is derived from the Greek root *ichthy*, meaning “fishy,” because this group of inherited disorders is associated with chronic, excessive keratin buildup (hyperkeratosis) that results in fish-like scaliness (Fig. 25.20A). Ichthyosis is subtyped according to the mode of inheritance, histology, and clinical features; the primary categories include *ichthyosis vulgaris* (autosomal dominant or acquired), *congenital ichthyosiform erythroderma* (autosomal

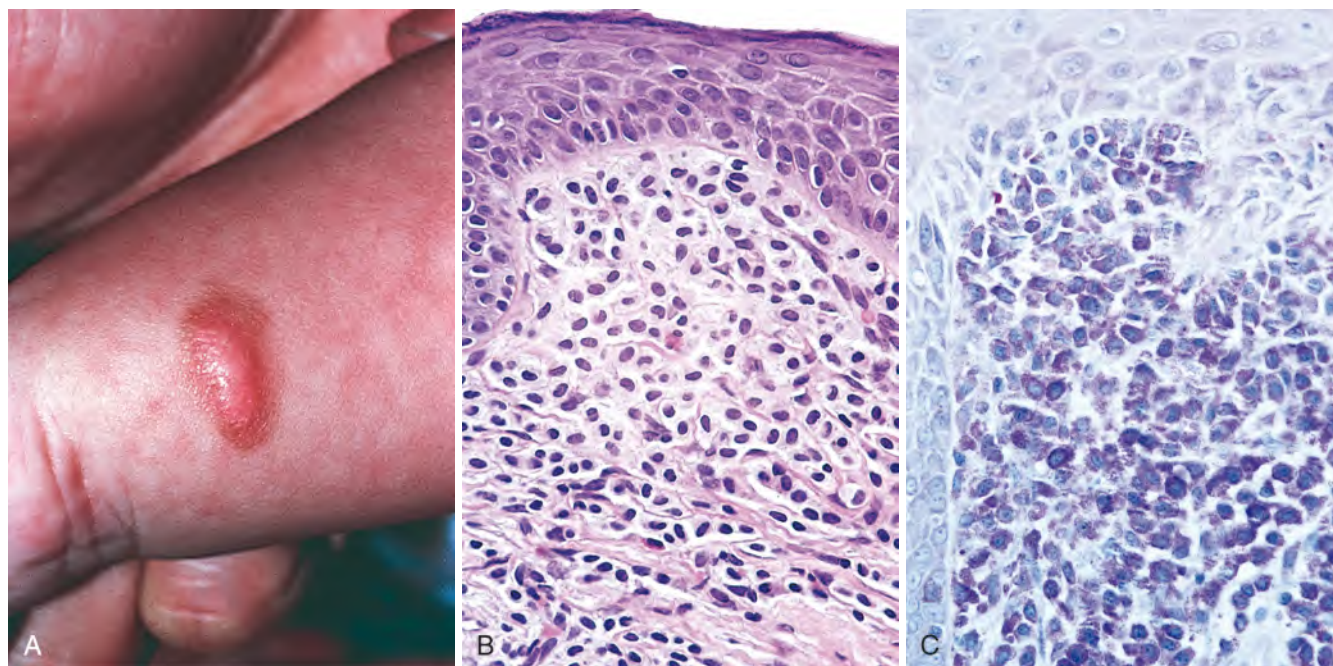


Fig. 25.19 Mastocytosis. (A) Solitary mastocytoma in a 1-year-old child. (B) By histology, numerous ovoid cells with uniform, centrally located nuclei are observed in the dermis. (C) Giemsa staining reveals purple “metachromatic” granules within the cytoplasm of the mast cells.

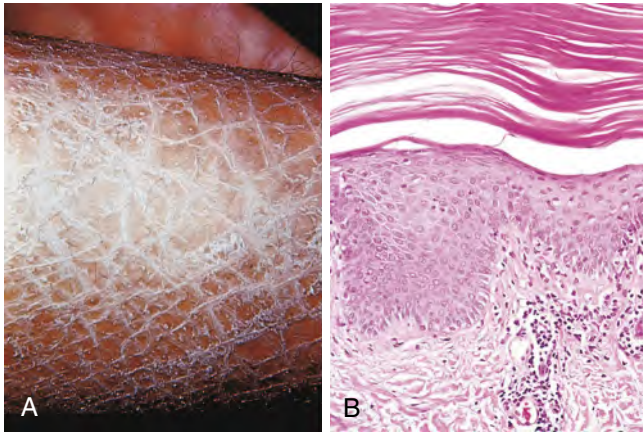


Fig. 25.20 Ichthyosis. Note prominent fish-like scales (A) and compacted, thickened stratum corneum (B).

recessive), *lamellar ichthyosis* (autosomal recessive), and *X-linked ichthyosis*. Most ichthyoses become apparent either at or around the time of birth. Acquired (noninherited) variants also exist and may be associated with lymphoid and visceral malignancies.

Pathogenesis

The primary abnormality in ichthyosis is defective desquamation, leading to retention of abnormally formed scale. For example, X-linked ichthyosis is caused by a deficiency of steroid sulfatase, an enzyme that helps to remove proadhesive cholesterol sulfate from intercellular spaces. In its absence, cholesterol sulfate accumulates, resulting in persistent cell-to-cell adhesion within the stratum corneum and a failure of desquamation.

MORPHOLOGY

All forms of ichthyosis exhibit a buildup of compacted and thickened stratum corneum that is associated with loss of the normal basket-weave pattern (Fig. 25.20B). There is generally little or no inflammation. Variations in the thickness of the epidermis and the stratum granulosum, the gross appearance, and the distribution of lesions are used to subclassify these disorders.

ACUTE INFLAMMATORY DERMATOSES

Innumerable inflammatory dermatoses have been described and can be broadly classified as acute and chronic. Acute lesions last from days to weeks and are characterized by inflammatory infiltrates (usually composed of lymphocytes and macrophages rather than neutrophils), edema, and variable degrees of epidermal, vascular, or subcutaneous injury. Chronic lesions, on the other hand, persist for months to years and are often associated with changes in epidermal growth (atrophy or hyperplasia) or dermal fibrosis. Discussed here are examples of the more commonly encountered acute dermatoses.

Urticaria

Urticaria (hives) is a common disorder of the skin that is usually caused by localized mast cell degranulation and is uniformly associated with dermal microvascular hyperpermeability. This combination of effects produces pruritic edematous plaques called *wheals*. Angioedema is closely related to urticaria and is characterized by edema of the deeper dermis and the subcutaneous fat.

Urticaria most often occurs between ages 20 and 40 years, but all age groups are susceptible. Individual lesions develop and fade within hours (usually less than 24 hours), and episodes may last for days or persist for months. Sites of predilection for urticarial eruptions include any area exposed to pressure, such as the trunk, distal extremities, and ears. Persistent episodes of urticaria may herald an underlying disease (e.g., collagen vascular disorders, Hodgkin lymphoma), but in most cases no underlying cause is identified.

Pathogenesis

Urticaria is most commonly the result of antigen-induced release of vasoactive mediators from mast cells. The various forms can be classified based on their dependence on IgE antibody and mast cells:

- *Mast cell-dependent, immunoglobulin E (IgE)-dependent.* Urticaria of this type follows exposure to many different antigens (pollens, foods, drugs, insect venom) and is an example of a localized immediate hypersensitivity (type I) reaction triggered by the binding of antigen to IgE antibodies that are attached to mast cells through Fc receptors (Chapter 6).
- *Mast cell-dependent, IgE-independent.* This subset results from substances that directly incite the degranulation of mast cells, such as opiates, certain antibiotics, and radiographic contrast media.
- *Mast cell-independent, IgE-independent.* These forms of urticaria are triggered by local factors that increase vascular permeability. One form is initiated by exposure to chemicals or drugs, such as aspirin, that inhibit cyclooxygenase and arachidonic acid production. The precise mechanism of aspirin-induced urticaria is unknown. A second form is *hereditary angioneurotic edema* (Chapter 6), caused by an inherited deficiency of C1 inhibitor that results in excessive activation of the early components of the complement system and production of vasoactive mediators.

MORPHOLOGY

Lesions vary from small, pruritic papules to large edematous plaques (Fig. 25.21A). Individual lesions may coalesce to form annular, linear, or arciform configurations. The histologic features of urticaria may be very subtle. There is usually a sparse superficial perivenular infiltrate consisting of mononuclear cells and rare neutrophils. Eosinophils may also be present. Collagen bundles are more widely spaced than in normal skin, a result of dermal edema (Fig. 25.21B). Dermal lymphatic channels may also be dilated due to increased absorption of edema fluid. There are no changes in the epidermis.

Acute Eczematous Dermatitis

The Greek word *eczema*, meaning “to boil over,” vividly describes the appearance of acute eczematous dermatitis—one of the most common skin disorders. Based on initiating factors, *eczematous dermatitis* can be subdivided into the following categories: (1) allergic contact dermatitis, (2) atopic dermatitis, (3) drug-related eczematous dermatitis, (4) photoeczematous dermatitis, and (5) primary irritant dermatitis.

The causes of eczema are sometimes broadly separated into “outside jobs” and “inside jobs”: disease resulting from external application of an antigen (e.g., poison ivy) or a reaction to an internal circulating antigen (which may be derived from ingested food or a drug). Treatment involves a search for offending substances that can be removed from the environment. Topical steroids can be used to block the inflammatory response. While such treatments are only palliative, they are nevertheless helpful in interrupting acute exacerbations of eczema, which can become self-perpetuating if unchecked.

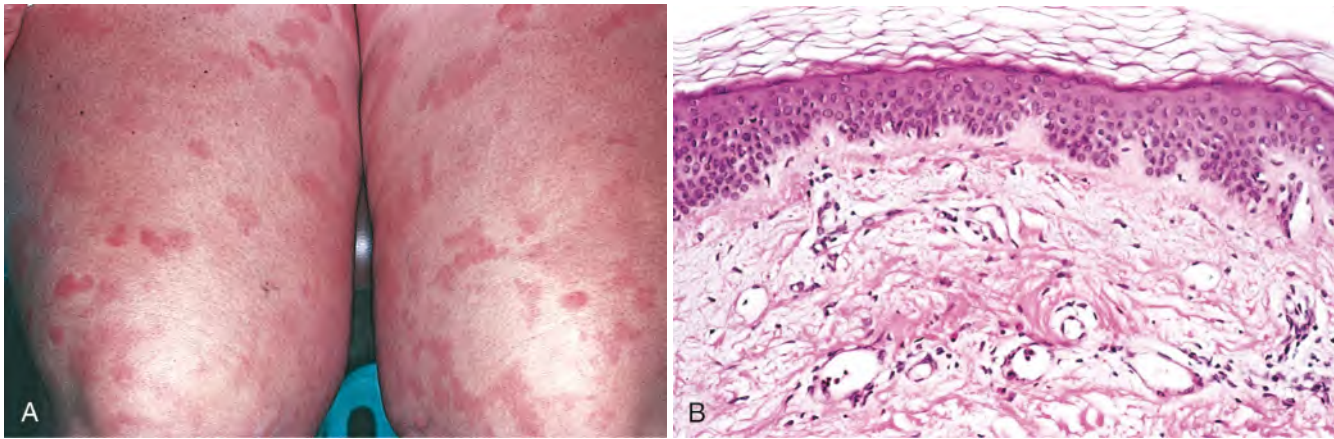


Fig. 25.21 Urticaria. (A) Erythematous, edematous, often circular plaques are characteristic. (B) Histologically, there is superficial dermal edema, manifested by spaces between collagen bundles, and dilated lymphatic and blood-filled vascular spaces; the epithelium is normal.

Pathogenesis

Eczematous dermatitis typically results from T cell-mediated inflammatory reactions (type IV hypersensitivity). This has been well studied in dermatitis triggered by contact antigens (e.g., urushiol from poison ivy). It is believed that chemicals rubbed onto the skin react with self proteins, thereby acting as “haptens” that create neoantigens. These antigens are taken up by Langerhans cells, which then migrate through dermal lymphatics to draining lymph nodes. Here they present the neoantigens to naive CD4⁺ T cells, which are activated and develop into effector and memory cells (Chapter 6). On antigen reexposure, memory T cells expressing homing molecules, such as common lymphocyte antigen and particular chemokine receptors, migrate to cutaneous sites where antigen is localized. Here they release cytokines and chemokines, which serve to recruit numerous additional inflammatory cells. This process occurs within 24 hours and accounts for the initial erythema and pruritus that characterize the acute, spongiotic phase of eczema.

Langerhans cells within the epidermis play a central role in contact dermatitis, and understandably factors that affect Langerhans cell function impact the inflammatory reaction. For example, chronic exposure to UV light is injurious to epidermal Langerhans cells and can prevent sensitization to contact antigens, but UV light can also alter antigens and generate forms that are more likely to induce sensitivity reactions.

MORPHOLOGY

A time course of the histologic changes associated with typical eczematous dermatitis is shown in Fig. 25.22. All types of eczematous dermatitis are characterized by red, papulovesicular, crusted lesions that, if persistent, develop reactive **acanthosis** and **hyperkeratosis** and take on the appearance of raised scaling plaques. A striking example of eczema is an acute contact reaction to topical antigens such as urushiol in poison ivy/oak (*Rhus toxicodendron*), which is characterized by pruritic, edematous, oozing plaques that often contain small and large blisters (vesicles and bullae) (Fig. 25.23A). Such lesions are prone to bacterial superinfection, which produces a yellow crust (impetiginization). With time, persistent lesions become less “wet” (fail to ooze or form vesicles) and become progressively more hyperkeratotic and acanthotic. **Spongiosis** characterizes acute eczematous dermatitis—hence the histologic synonym spongiotic dermatitis. Unlike urticaria, in which edema is restricted to the superficial

dermis, edema seeps into the intercellular spaces of the epidermis, splaying apart keratinocytes, particularly in the stratum spinosum. Mechanical shearing of intercellular attachment sites (desmosomes) and cell membranes by progressive accumulation of intercellular fluid may result in the formation of intraepidermal vesicles (Fig. 25.23B).

During the earliest stages of eczematous dermatitis, there is a superficial, perivascular, lymphocytic infiltrate associated with papillary dermal edema and mast cell degranulation. The pattern and composition of this infiltrate may provide clues to the underlying cause. For example, eczema resulting from certain ingested drugs is marked by perivascular infiltrates that often contain eosinophils in the superficial and deep dermis. By contrast, eczematous dermatitis resulting from contact antigens tends to produce a mononuclear inflammatory reaction without eosinophils that preferentially affects the superficial dermis.

Treatment of eczema revolves around avoiding further exposure to the inciting agent and using antiinflammatory drugs. Mild to moderate disease is typically treated with topical antiinflammatory agents such as corticosteroids, whereas severe disease often responds to biologic agents that block the action of cytokines, such as IL-4 and IL-13, that are produced by CD4⁺ Th2 T cells.

Erythema Multiforme

Erythema multiforme is an uncommon self-limited hypersensitivity reaction to certain infections and drugs. It affects individuals of any age and is associated with: (1) infections such as herpes simplex, mycoplasma, histoplasmosis, coccidioidomycosis, typhoid, and leprosy, among others; (2) exposure to certain drugs (sulfonamides, penicillin, barbiturates, salicylates, hydantoins, and antimalarials); (3) cancer (carcinomas and lymphomas); and (4) collagen vascular diseases (lupus erythematosus, dermatomyositis, and polyarteritis nodosa).

Pathogenesis

Erythema multiforme is characterized by keratinocyte injury mediated by skin-homing CD8⁺ cytotoxic T lymphocytes. This mechanism of injury is shared with several other conditions, including acute graft-versus-host disease, skin allograft rejection, and fixed drug eruptions. The epidermal antigens that are recognized by the infiltrating T cells in erythema multiforme remain unknown.

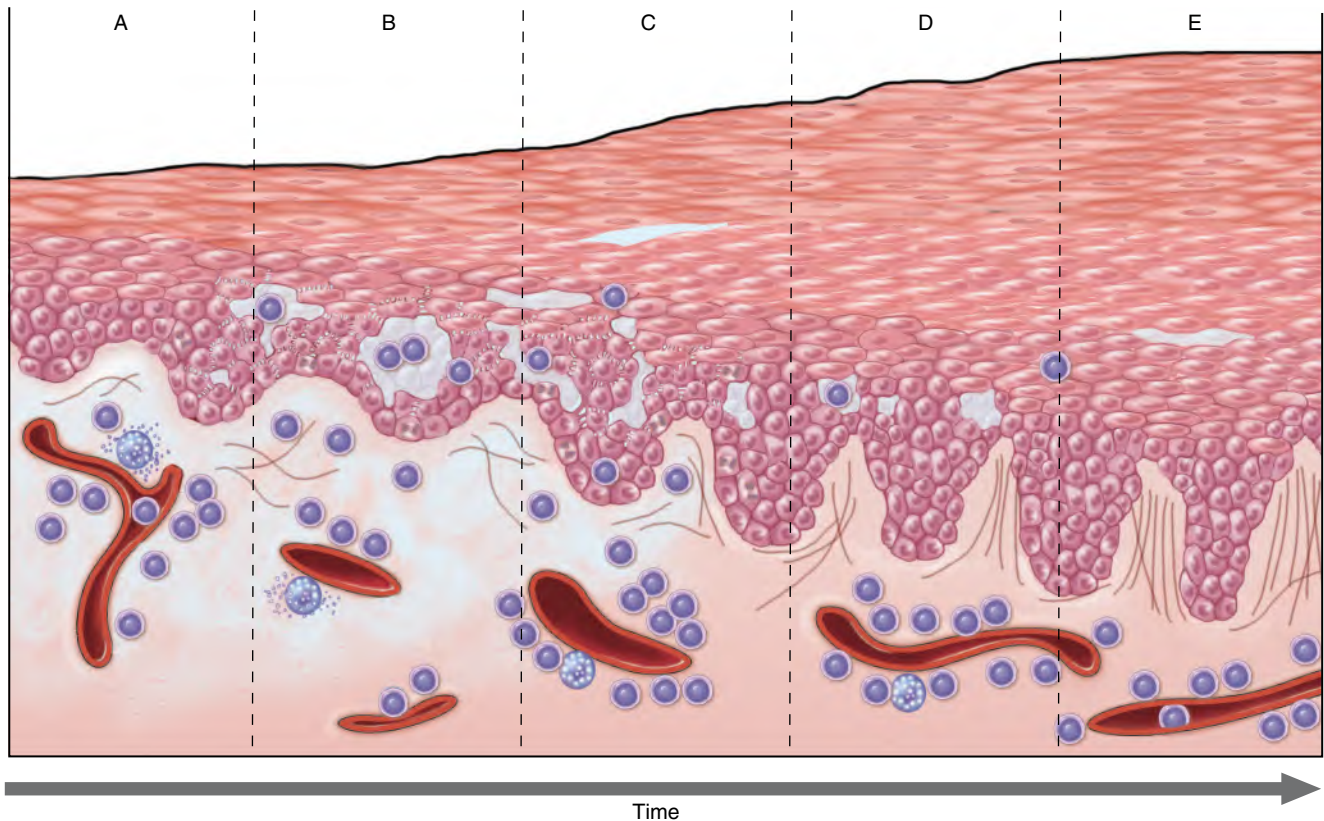


Fig. 25.22 Stages of eczema development. (A) Initial dermal edema and perivascular infiltration by inflammatory cells is followed within 24 to 48 hours by (B) epidermal spongiosis and microvesicle formation. (C) Abnormal scale, including parakeratosis, as well as progressive acanthosis (D) and hyperkeratosis (E) appear as the lesion becomes chronic.

MORPHOLOGY

Affected individuals present with a diverse array of lesions (hence the term *multiforme*), including macules, papules, vesicles, bullae, and characteristic targetoid (target-like) lesions (Fig. 25.24A), which may be less prominent in individuals with darker skin (Fig. 25.24B). The lesions may occur in a variety of distributions. Cases that are limited in extent often show symmetric involvement of the extremities. A febrile form associated with extensive involvement of the skin is called **Stevens-Johnson syndrome**, which is often (but not exclusively) seen in children. In Stevens-Johnson syndrome, lesions involve not only the skin but also the lips and oral mucosa, conjunctiva, urethra, and genital and perianal areas. Secondary infection of involved areas due to loss of skin integrity may result in life-threatening sepsis. Another variant termed **toxic epidermal necrolysis** is characterized by diffuse necrosis and sloughing of cutaneous and mucosal epithelial surfaces. The widespread epidermal damage produces a clinical picture similar to that seen in patients with extensive burns.

On histologic examination, the “targetoid” lesions show a superficial perivascular, lymphocytic infiltrate associated with dermal edema and accumulation of lymphocytes along the dermoepidermal junction, where they are intimately associated with degenerating and necrotic keratinocytes, a pattern termed *interface dermatitis* (Fig. 25.24C). With time there is upward migration of lymphocytes into the epidermis. Discrete and confluent zones of epidermal necrosis occur with concomitant blister formation. Epidermal sloughing leads to shallow erosions.

CHRONIC INFLAMMATORY DERMATOSES

This category includes inflammatory skin disorders that persist for months to years. The skin surface in chronic inflammatory dermatoses

is often roughened due to excessive or abnormal scale formation and diminished shedding.

Psoriasis

Psoriasis is a chronic inflammatory dermatosis that appears to have an autoimmune basis. It is a common disorder, affecting as many as 1% to 2% of people in the United States. Persons of all ages develop the disease. Approximately 15% of patients with psoriasis have associated arthritis, which may be mild or may produce severe deformities resembling the joint changes seen in rheumatoid arthritis. It can affect any joint in the body and may be symmetric or asymmetric. In addition, psoriasis may be associated with myopathy, enteropathy, and acquired immunodeficiency syndrome (AIDS).

Pathogenesis

As with other suspected disorders of autoimmunity, psoriasis is believed to be the product of environmental and genetic factors, including particular HLA gene variants. The culprit antigens remain elusive, but it appears that sensitized populations of CD4⁺ Th1 and Th17 cells and activated CD8⁺ cytotoxic effector T cells enter the skin and accumulate in the epidermis. This immune response provokes the secretion of factors that induce keratinocyte proliferation, resulting in the characteristic lesions. The interactions between CD4⁺ T cells, CD8⁺ T cells, dendritic cells, and keratinocytes give rise to a cytokine “soup” dominated by Th1-type and Th17-type cytokines such as IL-12, interferon- γ , tumor necrosis factor (TNF), and IL-17. The importance of these factors is highlighted by the generally



Fig. 25.23 Eczematous dermatitis. (A) Patterned erythema and scale stemming from a nickel-induced contact dermatitis produced by a necklace in an individual with lighter skin. (B) By contrast, eczematous dermatitis can appear hyperpigmented in individuals with darker skin, as in this case of atopic dermatitis. In other cases, postinflammatory hypopigmentation may be seen in darker skin types (not shown). (C) Microscopically, there is an accumulation of fluid (spongiosis) between epidermal cells, which may progress to blister formation. (B, From Poladian K, De Souza B, McMichael AJ. Atopic dermatitis in adolescents with skin of color. *Cutis*. 2019;104:164–168.)

excellent clinical responses that are observed in patients treated with TNF and IL-17 inhibitors. Lymphocytes also produce growth factors for keratinocytes that may contribute to epidermal thickening. Psoriatic lesions can be induced in susceptible individuals by local trauma, a process known as the *Koebner phenomenon*, presumably because trauma sets in motion a local inflammatory response that becomes self-perpetuating.

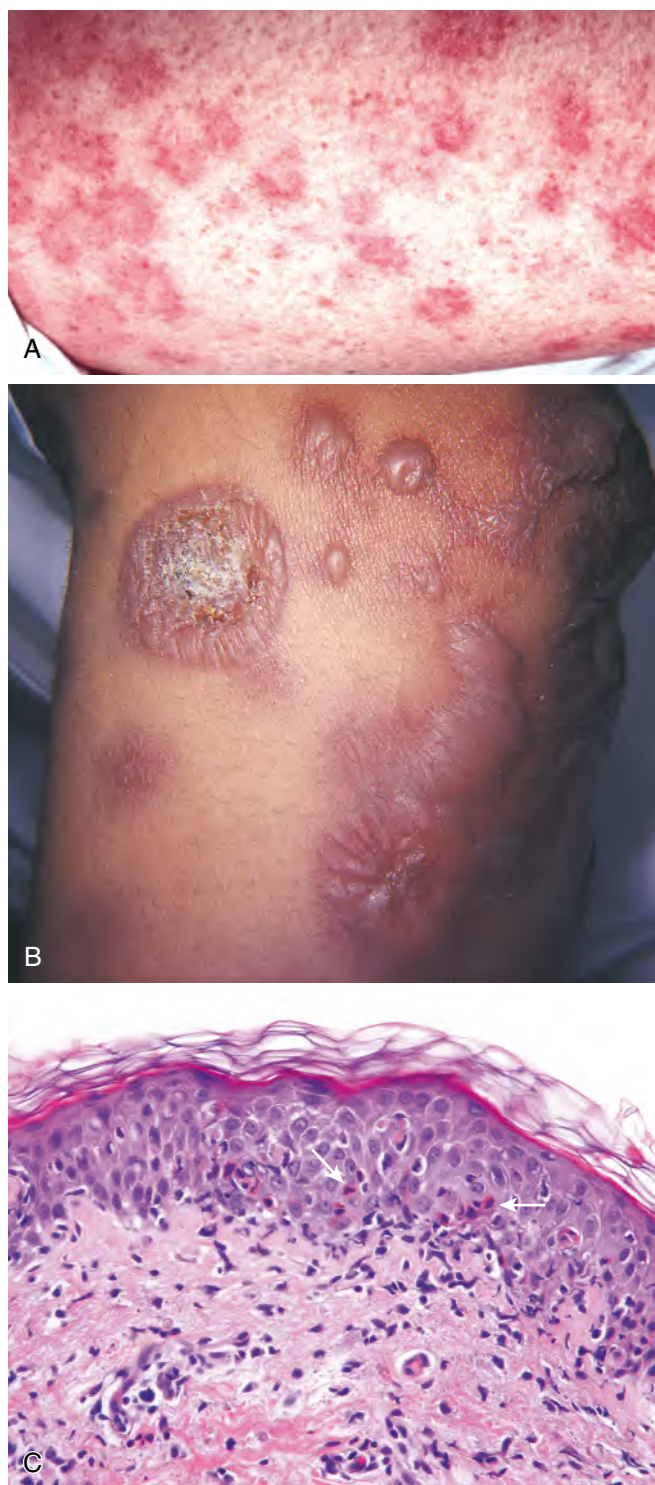


Fig. 25.24 Erythema multiforme. (A) On more lightly pigmented skin, well-demarcated, erythematous targetoid plaques are the hallmark of erythema multiforme. (B) On more darkly pigmented skin, the lesions may appear to be hyperpigmented and to have gray-silver central areas. (C) Early lesions show lymphocytes along the dermoepidermal junction (interface dermatitis) associated with scattered apoptotic keratinocytes (arrows), marked by dark shrunken nuclei and eosinophilic cytoplasm. (B, From Ibsen OAC, Peters SM. Immunity and immunologic oral lesions. In: *Ibsen OAC, Peters S. Oral Pathology for the Dental Hygienists, with General Pathology Introductions*. 8th ed. Elsevier; 2023.)

MORPHOLOGY

Psoriasis most frequently affects the skin of the elbows, knees, scalp, lumbosacral areas, intergluteal cleft, and glans penis. The typical lesion is a well-demarcated, **pink to salmon-colored plaque covered by loosely adherent silver-white scale** (Fig. 25.25A,B). Variations exist, with some lesions occurring in annular, linear, gyrate, or serpiginous configurations. Psoriasis is one cause of total body erythema and scaling known as erythroderma. **Nail changes** occur in 30% of cases of psoriasis and consist of yellow-brown discoloration (often likened to an oil slick), with pitting, dimpling, separation of the nail plate from the underlying bed (onycholysis), thickening, and crumbling.

Established lesions of psoriasis have a characteristic histologic picture. Increased epidermal cell proliferation results in marked epidermal thickening (acanthosis), with regular downward elongation of the rete ridges sometimes described as appearing like test tubes in a rack (Fig. 25.25C). Mitotic figures are easily identified well above the basal cell layer, where mitotic activity is confined in normal skin. **The stratum granulosum is thinned or absent and extensive overlying parakeratotic scale is seen.** Typical of psoriatic plaques is thinning of the portion of the epidermal cell layer that overlies the tips of the dermal papillae (suprapapillary plates), which contain dilated, tortuous blood vessels. The abnormal proximity of vessels within the dermal papillae to the overlying parakeratotic scale accounts for the characteristic clinical phenomenon of multiple, minute, bleeding points when the scale is lifted from the plaque (**Auspitz sign**). Neutrophils form small aggregates within the superficial epidermis (**spongiform pustules**) and the parakeratotic stratum corneum (**Munro microabscesses**). In pustular psoriasis, larger abscess-like accumulations of neutrophils are present directly beneath the stratum corneum.

Seborrheic Dermatitis

Seborrheic dermatitis is a chronic inflammatory dermatosis that is even more common than psoriasis, affecting up to 5% of the general population. It classically involves regions with a high density of sebaceous glands, such as the scalp, forehead (especially the glabella), external auditory canal, retroauricular area, nasolabial folds, and presternal area. Despite this association and its name, seborrheic dermatitis is associated with inflammation of the epidermis and is not a disease of the sebaceous glands per se.

Pathogenesis

The precise etiology of seborrheic dermatitis is unknown. Increased sebum production, often in response to androgens, is one possible contributory factor. Involvement of sebum is supported by clinical observations of patients with Parkinson disease, who typically show increased sebum production secondary to dopamine deficiency and have a markedly increased incidence of seborrheic dermatitis. Treatment with levodopa decreases the oiliness of the skin and improves the seborrheic dermatitis. However, other conditions with increased sebum production such as acne (discussed later) are not associated with seborrheic dermatitis, and sebum production is probably best viewed as being necessary but not sufficient to cause the disorder. The composition of sebum, integrity of the epithelial barrier, and the host immune response have also been implicated. Other work has suggested a relationship with colonization of the skin by certain fungal species of the genus *Malassezia*, which may produce a lipase that releases proinflammatory free fatty acids. A severe form of seborrheic dermatitis that is difficult to treat was once common in many human immunodeficiency virus (HIV)-infected individuals with low CD4 counts, but its incidence has fallen with the advent of effective antiviral therapy.

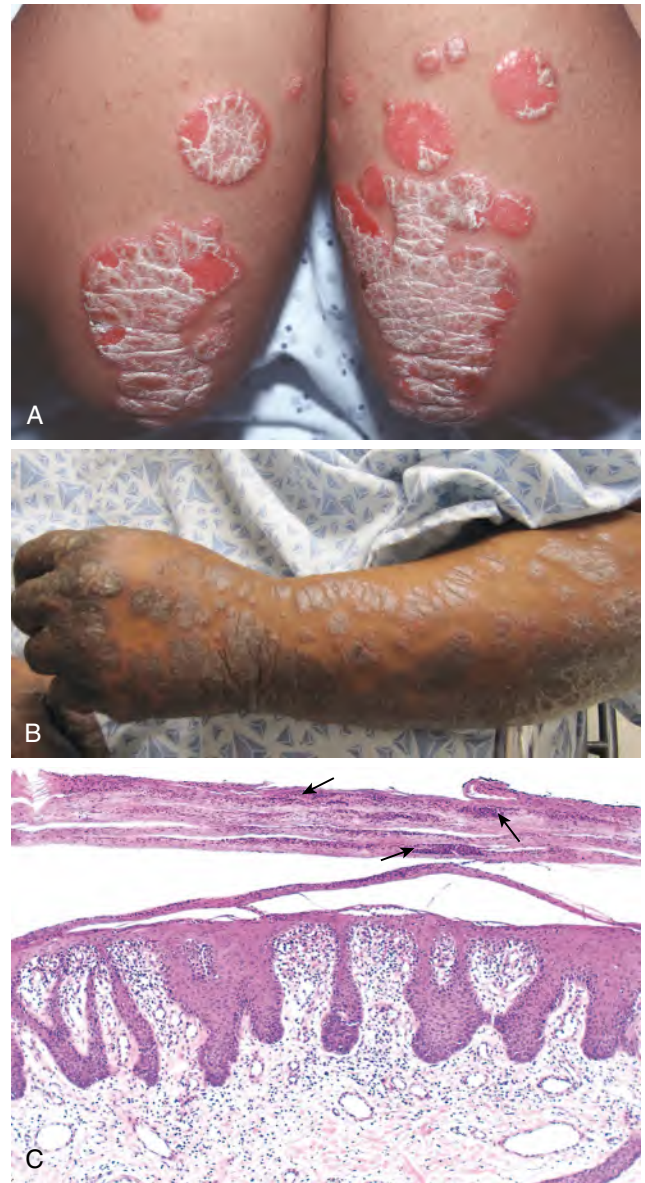


Fig. 25.25 Psoriasis. (A) Erythematous psoriatic plaques covered by silvery-white scale in an individual with lighter skin. (B) In patients with darker skin, plaques range from salmon-colored to hyperpigmented and typically demonstrate gray scale. (C) Microscopic examination shows marked epidermal hyperplasia, downward extension of rete ridges (psoriasiform hyperplasia), and prominent parakeratotic scale with infiltrating neutrophils (Munro microabscesses, arrows). (B, Courtesy Dr. Sarah Wolfe, Department of Dermatology, Duke University Medical Center, Durham, NC.)

MORPHOLOGY

The individual lesions are macules and papules on an erythematous-yellow, often greasy base, typically in association with extensive scaling and crusting. Fissures may also be present, particularly behind the ears. Dandruff is the common clinical expression of seborrheic dermatitis of the scalp. Microscopically, seborrheic dermatitis shares features with spongiotic dermatitis and psoriasis, with earlier lesions being more spongiotic and later ones more acanthotic. Typically, mounds of parakeratosis containing neutrophils and serum are present at the ostia of hair follicles (so-called **follicular lipping**). A superficial perivascular inflammatory infiltrate generally consists of a mixture of lymphocytes and neutrophils.

Lichen Planus

“Pruritic, purple, polygonal, planar, papules, and plaques” are the tongue-twisting “six Ps” of lichen planus, a disorder of skin and mucosa. Lichen planus is usually self-limited, most commonly resolving spontaneously 1 to 2 years after onset. Resolution often leaves a residuum of postinflammatory hyperpigmentation. Oral lesions, however, may persist for years. Squamous cell carcinoma sometimes arises from chronic mucosal and paramucosal lichen planus and is an example of carcinogenesis fostered by chronic inflammation. As in psoriasis, the *Koebner phenomenon* may be seen in lichen planus.

Pathogenesis

The pathogenesis of lichen planus is not known. It is possible that expression of altered antigens in basal epidermal cells or the dermoepidermal junction elicits a cell-mediated cytotoxic (CD8+) T-cell response.

MORPHOLOGY

Cutaneous lesions consist of itchy, violaceous, flat-topped papules that may coalesce to form plaques (Fig. 25.26A). The papules are often highlighted by white dots or lines called **Wickham striae**, which are created by areas of hypergranulosis. In darkly pigmented individuals, lesions may acquire a dark brown color due to release of melanin into the dermis as the basal cell layer is destroyed. Lesions are usually multiple and symmetrically distributed, particularly on the extremities and about the wrists and elbows. The glans penis is another common site of involvement. In 70% of cases, oral mucosal lesions are present that have a white, reticulated, or net-like appearance.

Lichen planus is characterized histologically by a dense, continuous infiltrate of lymphocytes along the dermoepidermal junction, a prototypic example of **interface dermatitis** (Fig. 25.26B). The lymphocytes are intimately associated with basal keratinocytes, which show degeneration, necrosis, and a resemblance in size and contour to more mature cells of the stratum spinosum (squamatization). Because of this destructive lymphocytic infiltrate, the dermoepidermal interface takes on an angulated zigzag contour (sawtoothing). Anucleate, necrotic basal cells may become incorporated into the inflamed papillary dermis, where they are referred to as **colloid** or **Civatte bodies**. Although the lesions bear some similarities to those in erythema multiforme, lichen planus shows changes of chronicity, namely, epidermal hyperplasia (or rarely atrophy) and thickening of the granular cell layer and stratum corneum (hypergranulosis and hyperkeratosis, respectively).

KEY CONCEPTS

Inflammatory Dermatoses

- Many specific inflammatory dermatoses exist, which can be mediated by IgE antibodies (urticaria) or antigen-specific T cells (eczema, erythema multiforme, and psoriasis).
- These disorders are diagnosed based on the distribution and gross appearance of skin lesions and the microscopic patterns of inflammation (e.g., interface dermatitis in lichen planus and erythema multiforme).

BLISTERING (BULLOUS) DISEASES

Although vesicles and bullae (blisters) occur in diverse unrelated conditions such as herpesvirus infection, erythema multiforme, and thermal burns, there exists a group of disorders in which blisters are the primary and most distinctive features. These *bullous diseases* produce dramatic lesions and in some instances are fatal if untreated. Blisters in the various disorders occur at different levels within the skin (Fig. 25.27); histologic assessment is essential for accurate diagnosis and provides insight into the pathogenic mechanisms. Knowledge of the structure of desmosomes and hemidesmosomes (described in Chapter 1), which you will recall provide the skin with mechanical stability, is helpful in understanding these diseases, as they are often caused by acquired or inherited defects in proteins that make up or bind to these structures (Fig. 25.28).

Inflammatory Blistering Disorders

Pemphigus

Pemphigus is a blistering disorder caused by autoantibodies that result in the dissolution of intercellular attachments within the epidermis and mucosal epithelium. The pathobiology of blistering disorders provides important insights into the molecular underpinnings of keratinocyte adhesion. Most individuals who develop pemphigus are in the fourth to sixth decades of life; males and females are affected equally. There are multiple variants: (1) *pemphigus vulgaris*, (2) *pemphigus vegetans*, (3) *pemphigus foliaceus*, (4) *pemphigus erythematosus*, and (5) *paraneoplastic pemphigus*.

- Pemphigus vulgaris*, by far the most common type (accounting for more than 80% of cases worldwide), involves the mucosa and

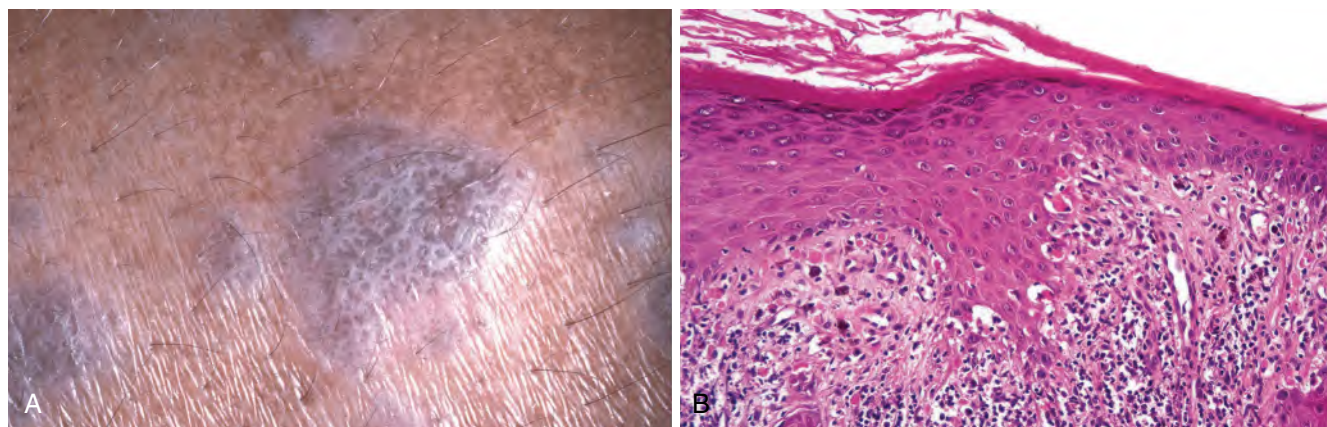


Fig. 25.26 Lichen planus. (A) This flat-topped pink-purple, polygonal papule has a white lace-like pattern of lines that are referred to as Wickham striae. (B) There is a band-like infiltrate of lymphocytes at the dermoepidermal junction, hyperkeratosis, and pointed rete ridges (sawtoothing), the last as a result of chronic basal cell layer injury.

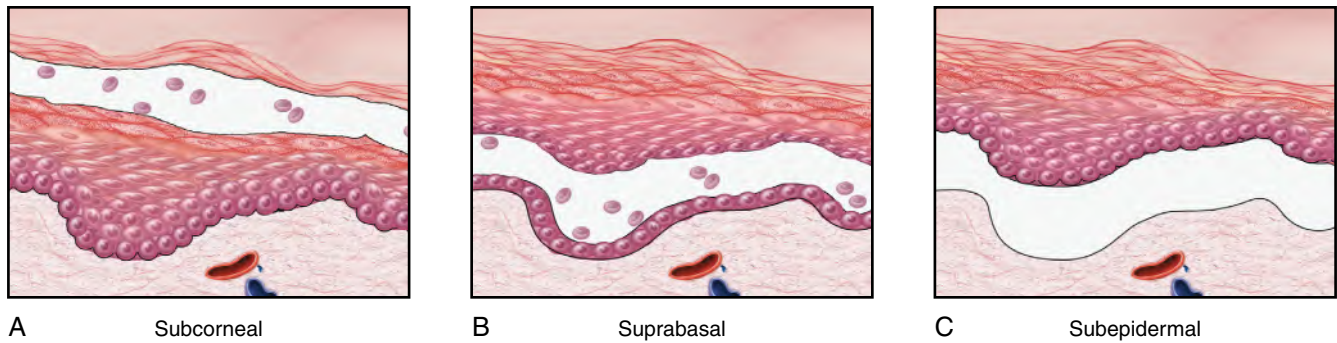


Fig. 25.27 Schematic representation of different types of blisters. (A) In subcorneal blisters, the stratum corneum forms the roof of the bulla (as in pemphigus foliaceus). (B) In a suprabasal blister, a portion of the epidermis, including the stratum corneum, forms the roof (as in pemphigus vulgaris). (C) In a subepidermal blister, the entire epidermis separates from the dermis (as in bullous pemphigoid).

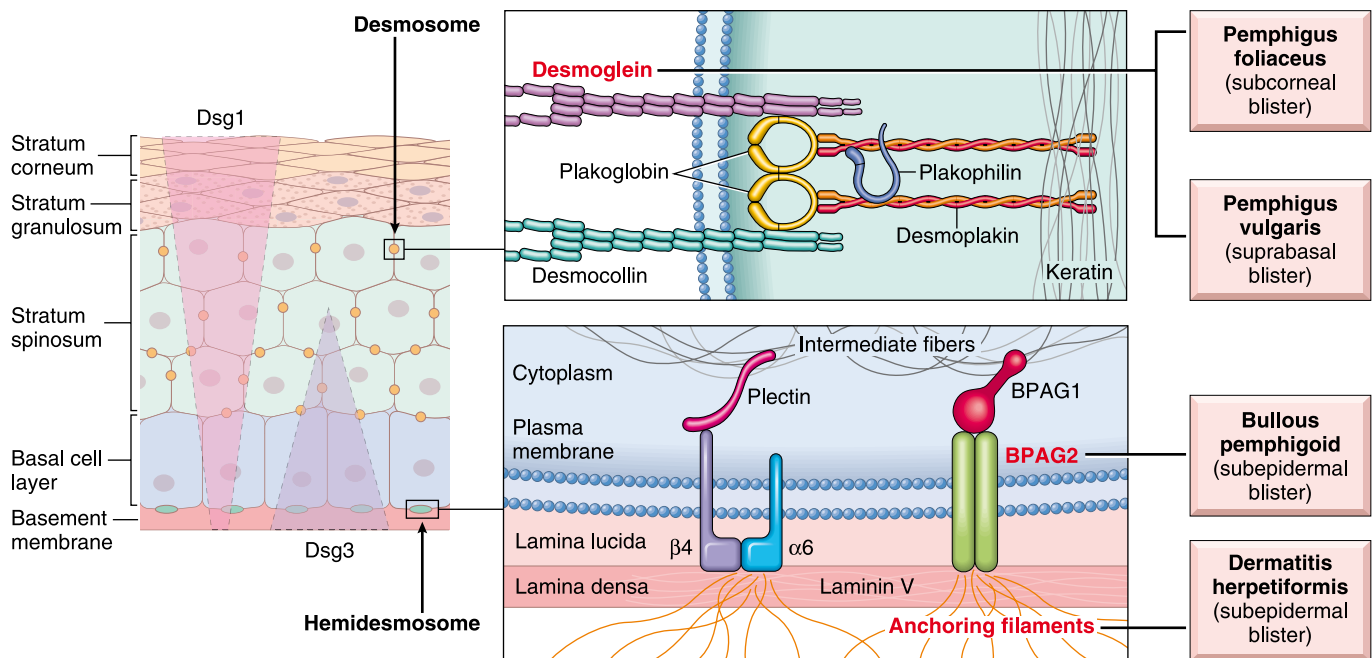


Fig. 25.28 Keratinocyte adhesion molecules and blistering inflammatory disorders. Knowledge of the proteins composing desmosomes and hemidesmosomes is key to understanding blistering disorders. Desmogleins 1 and 3 (*Dsg1*, *Dsg3*) are functionally interchangeable components of desmosomes but have different distributions within the epidermis (*left panel*), with *Dsg1* expression rising with keratinocyte differentiation and *Dsg3* expression falling. The major structural proteins of desmosomes and hemidesmosomes are shown in the *right panel*. In pemphigus vulgaris, autoantibodies against *Dsg1* and *Dsg3* cause blisters in the deep suprabasal epidermis, whereas in pemphigus foliaceus, autoantibodies against *Dsg1* alone lead to superficial, subcorneal blisters. In bullous pemphigoid, autoantibodies bind BPAG2, a component of the hemidesmosomes, leading to blister formation at the level of the lamina lucida of the basement membrane. Dermatitis herpetiformis is caused by IgA autoantibodies to the fibrils that anchor hemidesmosomes to the dermis.

skin, especially on the scalp, face, axilla, groin, trunk, and points of pressure. It may present as oral ulcers that persist for months before skin involvement appears. Primary lesions are superficial vesicles and bullae that rupture easily, leaving shallow erosions covered with dried serum and crust (**Fig. 25.29A**). It is usually not severe, but extreme cases can be fatal without treatment.

- *Pemphigus vegetans* is a rare form that usually presents not with blisters but with large, moist, verrucous (wart-like), vegetating

plaques studded with pustules on the groin, axillae, and flexural surfaces.

- *Pemphigus foliaceus* is a mild form that is endemic in Brazil (where it is called *fogo selvagem*) and occurs sporadically in other geographic regions. Lesions are most common on the scalp, face, chest, and back; the mucous membranes are only rarely affected. Bullae are so superficial that they mainly present as areas of erythema and crusting; these represent superficial erosions at sites of blister rupture (**Fig. 25.30A**).

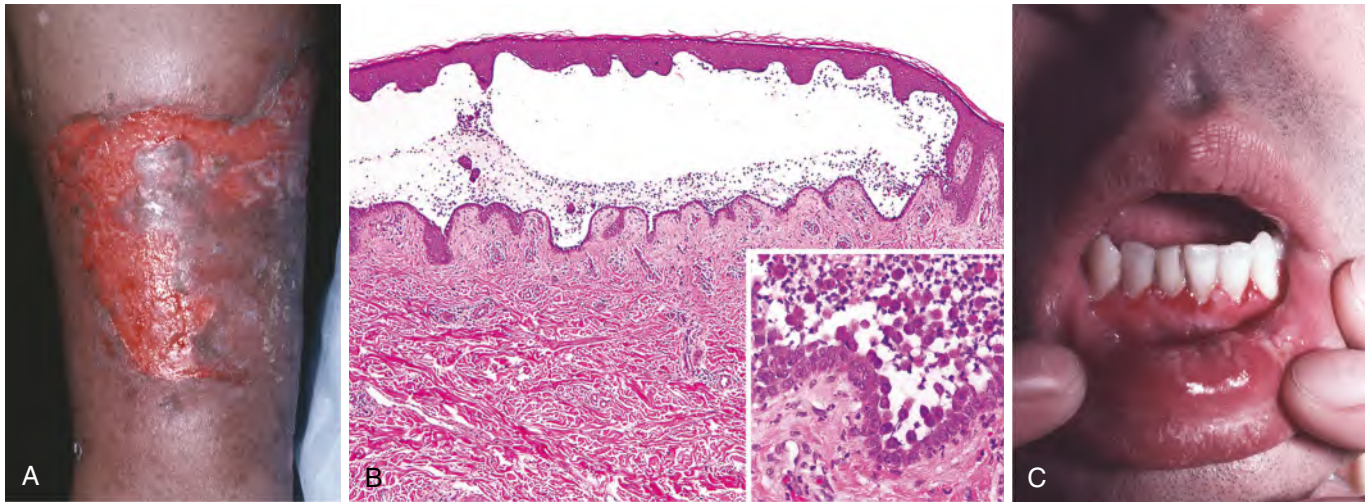


Fig. 25.29 Pemphigus vulgaris. (A) Eroded plaques are formed following the rupture of confluent, thin-roofed bullae. (B) Suprabasal acantholysis results in an intraepidermal blister in which dyscohesive (acantholytic) epidermal cells are present (*inset*). (C) Ulcerated blisters in the oral mucosa are also common, as seen here on the lip.

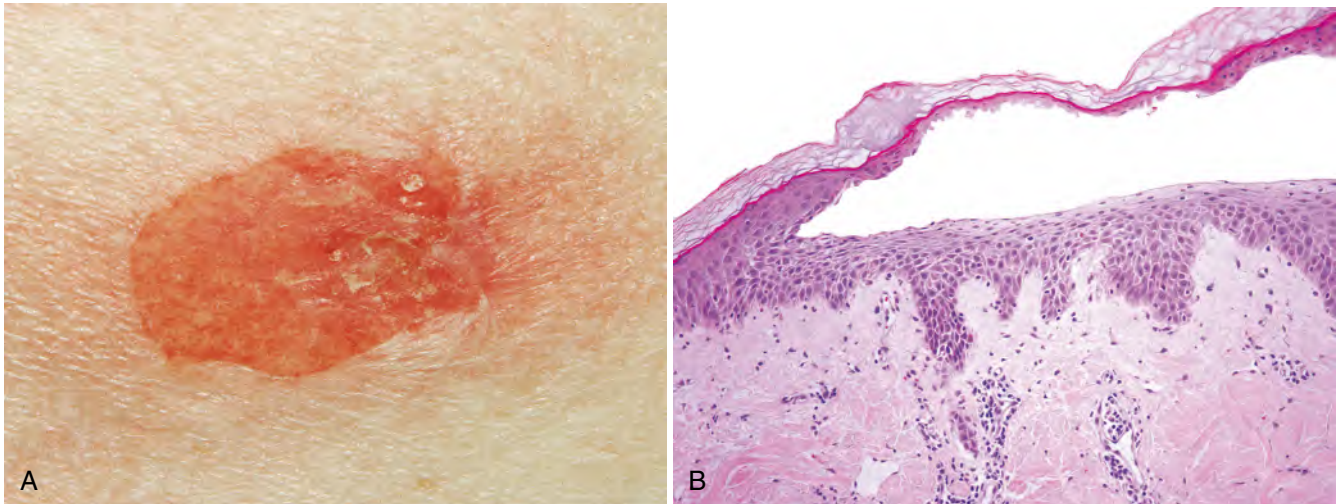


Fig. 25.30 Pemphigus foliaceus. (A) The delicate superficial (subcorneal) blisters are much less erosive than those seen in pemphigus vulgaris. (B) Subcorneal separation of the epithelium is seen microscopically.

- *Pemphigus erythematosus* is a localized, mild form of pemphigus foliaceus that selectively involves the malar area of the face in a lupus erythematosus-like fashion.
- *Paraneoplastic pemphigus* occurs in association with various malignancies, most commonly non-Hodgkin lymphoma.

Pathogenesis

All forms of pemphigus are autoimmune diseases caused by IgG autoantibodies against desmogleins that disrupt desmosome integrity and result in the formation of blisters. The distribution of desmoglein 1 and 3 in the epidermis and whether there are autoantibodies to one or both proteins appear to explain the position and severity of the blisters (see Fig. 25.28). In addition to directly disrupting the intercellular adhesive function of the desmosomes, the pathogenic antibodies may also act indirectly by activating intercellular proteases. Paraneoplastic pemphigus arises most often in the setting of lymphoid neoplasms and is also caused by autoantibodies that recognize desmogleins or other proteins involved in intercellular adhesion.

MORPHOLOGY

The common histologic feature in all forms of pemphigus is **acantholysis**, the dissolution of the intercellular bridges that connect squamous epithelial cells. Acantholytic cells dissociate from one another, lose their polyhedral shape, and become rounded. In pemphigus vulgaris (see Fig. 25.29B) and pemphigus vegetans, acantholysis selectively involves the cells immediately above the basal cell layer, producing a suprabasal blister. In the vegetans variant, there is also overlying epidermal hyperplasia. The single layer of intact basal cells that forms the blister base in pemphigus vulgaris has been likened to a row of tombstones. In pemphigus foliaceus, blisters form by similar mechanisms but are found in the superficial epidermis at the level of the stratum granulosum (see Fig. 25.30B). Variable superficial dermal infiltration by lymphocytes, macrophages, and eosinophils accompanies each type of pemphigus. By direct immunofluorescence, lesions show a characteristic net-like pattern of intercellular IgG deposits. IgG is usually seen at all levels of the epithelium in pemphigus vulgaris but tends to be more superficial in pemphigus foliaceus (Fig. 25.31).

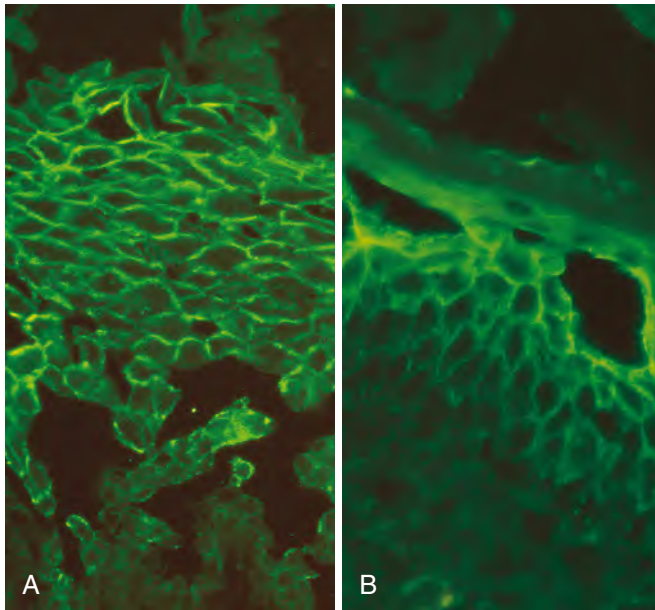


Fig. 25.31 Direct immunofluorescence staining for immunoglobulin of epidermis involved by pemphigus. (A) In pemphigus vulgaris there is deposition of immunoglobulin along the plasma membranes of keratinocytes in a reticular or fishnet-like pattern accompanied by suprabasilar loss of cell-to-cell adhesion (acantholysis). (B) In pemphigus foliaceus the immunoglobulin deposits and acantholysis are more superficial.

The mainstay of treatment in all forms of pemphigus is immunosuppressive agents, which decrease the titers of the pathogenic antibodies.

Bullous Pemphigoid

Generally affecting older individuals, bullous pemphigoid shows a wide range of clinical presentations. Sites of involvement include the inner aspects of the thighs, flexor surfaces of the forearms, axillae, groin, and lower abdomen. Oral lesions are present in 10% to 15% of affected individuals, usually appearing after the cutaneous lesions. Some patients present with urticarial plaques and severe pruritus.

Pathogenesis

Bullous pemphigoid is caused by autoantibodies that bind to proteins that are required for adherence of basal keratinocytes to the basement membrane. The so-called bullous pemphigoid antigens (BPAGs) are components of hemidesmosomes (see Fig. 25.28). Antibodies against one such component called BPAG2 are proven to cause blistering. Pathogenic autoantibodies also activate complement, leading to the recruitment of neutrophils and eosinophils, inflammation, and disruption of epidermal attachments.

MORPHOLOGY

The lesions are tense bullae filled with clear fluid involving erythematous or normal-appearing skin (Fig. 25.32A). The bullae are usually less than 2 cm in diameter but occasionally may reach 4 to 8 cm in diameter. They do not rupture easily, unlike the blisters seen in pemphigus, and heal without scarring unless they become infected secondarily. The separation of bullous pemphigoid from pemphigus is based on the identification of **subepidermal, nonacantholytic** blisters. Early lesions show a superficial and sometimes deep perivascular infiltrate of lymphocytes and variable numbers of eosinophils, occasional neutrophils, superficial dermal edema, and basal cell layer vacuolization (Fig. 25.32B). Eosinophils are typically present directly beneath the epidermal basal cell layer. The vacuolated basal cell layer eventually lifts away, allowing space for a fluid-filled blister to form. Most antibody deposition occurs in a continuous

linear pattern at the dermoepidermal junction (Fig. 25.32C), where specialized structures called hemidesmosomes link basal keratinocytes to the underlying basement membrane (Fig. 25.32D).

Dermatitis Herpetiformis

Dermatitis herpetiformis is a rare disorder characterized by urticaria and grouped vesicles. The disease affects predominantly males, most often in the third and fourth decades of life. Individuals with celiac disease have a greater than 100-fold increased risk of dermatitis herpetiformis. The plaques and vesicles are extremely pruritic.

Pathogenesis

Dermatitis herpetiformis is caused by IgA antibodies that cross-react with reticulin, a component of the anchoring fibrils that tether the epidermal basement membrane to the superficial dermis. These antibodies often appear in individuals who are genetically predisposed to develop antibodies to dietary gluten (derived from the wheat protein gliadin), explaining the association with celiac disease (Chapter 17). The resultant injury and inflammation produce a subepidermal blister. In some people with dermatitis herpetiformis and gluten-sensitive enteropathy, both disorders respond to a gluten-free diet.

MORPHOLOGY

The lesions are bilateral, symmetric, and grouped, and preferentially involve the extensor surfaces, elbows, knees, upper back, and buttocks (Fig. 25.33A). The lesions may appear to be hyperpigmented with they occur in individuals with darker skin (Fig. 25.33B). Fibrin and neutrophils accumulate selectively at the **tips of dermal papillae**, forming small microabscesses (Fig. 25.33C). The basal cells overlying these microabscesses show vacuolization and focal dermoepidermal separation that ultimately coalesce to form a true **subepidermal blister**. By direct immunofluorescence, dermatitis herpetiformis shows discontinuous **granular deposits of IgA** that selectively localize in the tips of dermal papillae (Fig. 25.33D).

Noninflammatory Blistering Disorders

Epidermolysis Bullosa and Porphyrria

Some disorders characterized by vesicles and bullae are mediated by inherited or, in some cases, acquired defects involving structural proteins that maintain the integrity of the skin. Two such disorders are epidermolysis bullosa and porphyria.

Epidermolysis Bullosa. Epidermolysis bullosa is a blanket term for a group of disorders caused by inherited defects in structural proteins that lend mechanical stability to the skin. The common feature is a proclivity to form blisters at sites of pressure, rubbing, or trauma at or soon after birth. The histologic changes in all forms are so subtle that electron microscopy may be required to differentiate among the various types.

- In the *simplex type*, defects of the basal cell layer of the epidermis almost always result from mutations in the genes encoding keratin 14 or keratin 5. These two proteins normally pair with one another to make a functional keratin fiber, thus explaining the similar phenotype resulting from mutations in either gene. The mutated proteins have a dominant negative activity and as a result the disorder shows an autosomal dominant mode of inheritance. This is the most common type of epidermolysis bullosa, encompassing 75% to 85% of cases.
- In the *junctional type*, blisters occur in otherwise histologically normal skin at the level of the lamina lucida (Fig. 25.34; see Fig. 25.28). Most cases are caused by autosomal recessive defects in one of the subunits of *laminin*, a multicomponent protein located in the lamina lucida that binds to both hemidesmosomes and anchoring filaments. Some of the remaining cases are caused by mutations in BPAG2, the same protein that is targeted by autoantibodies in bullous pemphigoid.

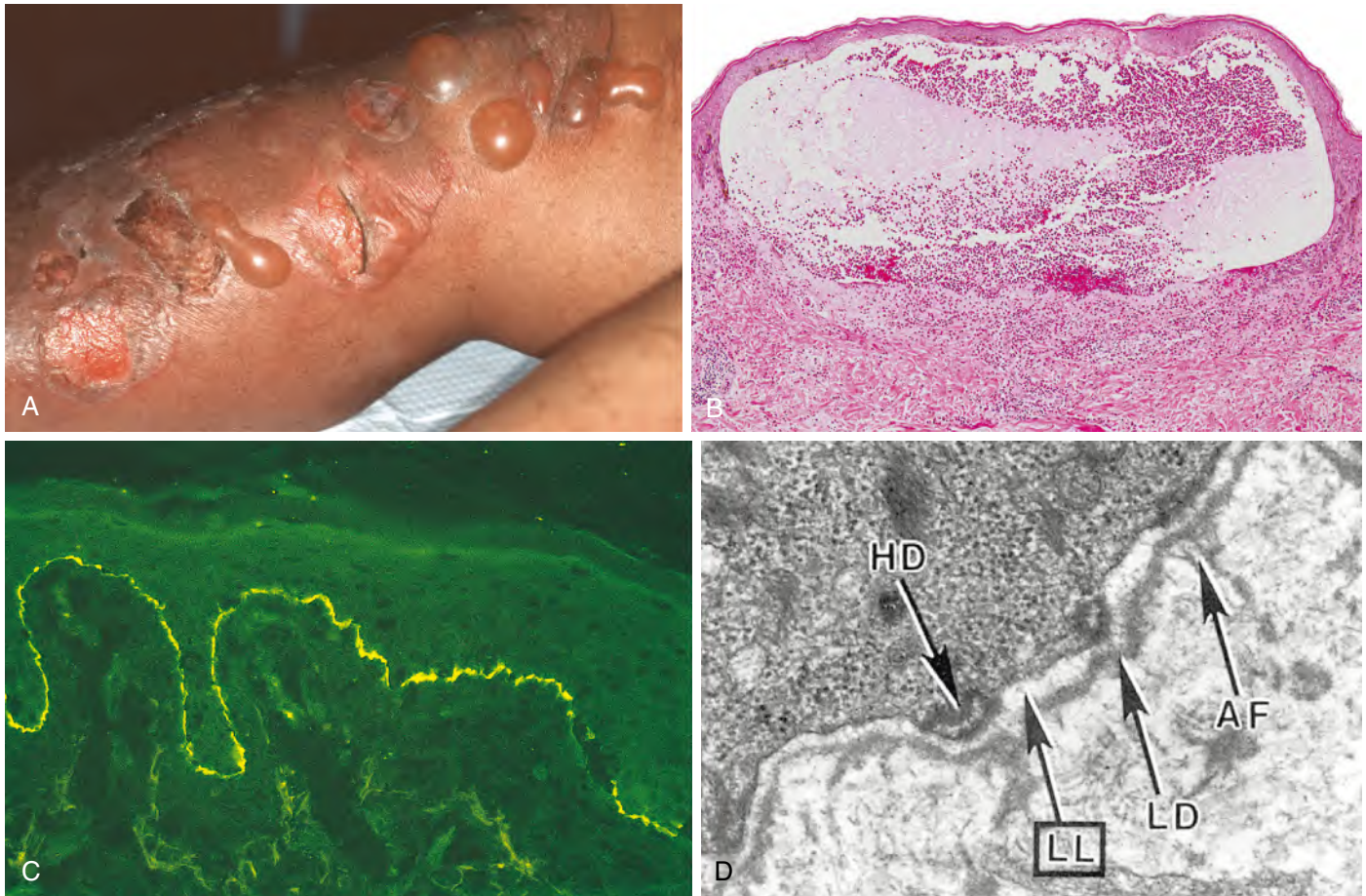


Fig. 25.32 Bullous pemphigoid. (A) Bullae consist of tense blisters that usually fail to rupture, as their roof consists of intact epidermis. Ulcers form if the blisters rupture. (B) Intact subepidermal blister associated with eosinophils, lymphocytes, and occasional neutrophils. (C) Linear deposition of complement along the dermoepidermal junction (see Fig. 25.31 for comparison). (D) Electron micrograph showing the ultrastructural features of the dermoepidermal junction. The bullous pemphigoid antigen (BPAG) is located in the basal portion of basal keratinocytes in association with hemidesmosomes (HD), which attach the epidermis to the lamina lucida (LL) of the basement membrane. AF, Anchoring fibrils; LD, lamina densa.

- In the scarring *dystrophic types*, blisters develop beneath the lamina densa in association with rudimentary or defective anchoring fibrils. Dystrophic epidermolysis bullosa usually results from mutations in the *COL7A1* gene, which encodes type VII collagen, a major component of the basement membrane anchoring fibrils. Depending on the mutation, the disorder may follow an autosomal dominant or autosomal recessive mode of inheritance.
- *Mixed types*, marked by defects at several levels, are also recognized.

Porphyria. Porphyria refers to a group of uncommon inborn or acquired disturbances of porphyrin metabolism. Porphyrins are pigments that are normally present in hemoglobin, myoglobin, and cytochromes. The classification of porphyria is based on both clinical and biochemical features. The five major types are (1) *congenital erythropoietic porphyria*, (2) *erythrohepatic protoporphyria*, (3) *acute intermittent porphyria*, (4) *porphyria cutanea tarda*, and (5) *mixed porphyria*. Cutaneous manifestations consist of urticaria and vesicles associated with scarring that are exacerbated by exposure to sunlight. The vesicles are subepidermal in location, and the adjacent dermis contains vessels with walls that are thickened by glassy deposits of serum proteins, including immunoglobulins (Fig. 25.35). The pathogenesis of these alterations is not understood.

KEY CONCEPTS

Blistering Disorders

- Blistering disorders are classified based on the level of epidermal separation.
- These disorders are often caused by autoantibodies specific for epithelial or basement membrane proteins that lead to unmooring of keratinocytes (acantholysis).
- Pemphigus is associated with IgG autoantibodies to various intercellular desmogleins, resulting in bullae that are either subcorneal (pemphigus foliaceus) or suprabasilar (pemphigus vulgaris).
- Bullous pemphigoid is associated with IgG autoantibodies to basement membrane proteins and produces a subepidermal blister.
- Dermatitis herpetiformis is associated with IgA autoantibodies that produces subepidermal blisters by binding to fibrils that join the epidermal basement membrane to the dermis.
- Noninflammatory blistering disorders include inherited defects in proteins that stabilize the epidermis (e.g., epidermolysis bullosa), as well as defects in porphyrin synthesis (porphyrias) that lead to sun-induced skin damage through uncertain mechanisms.

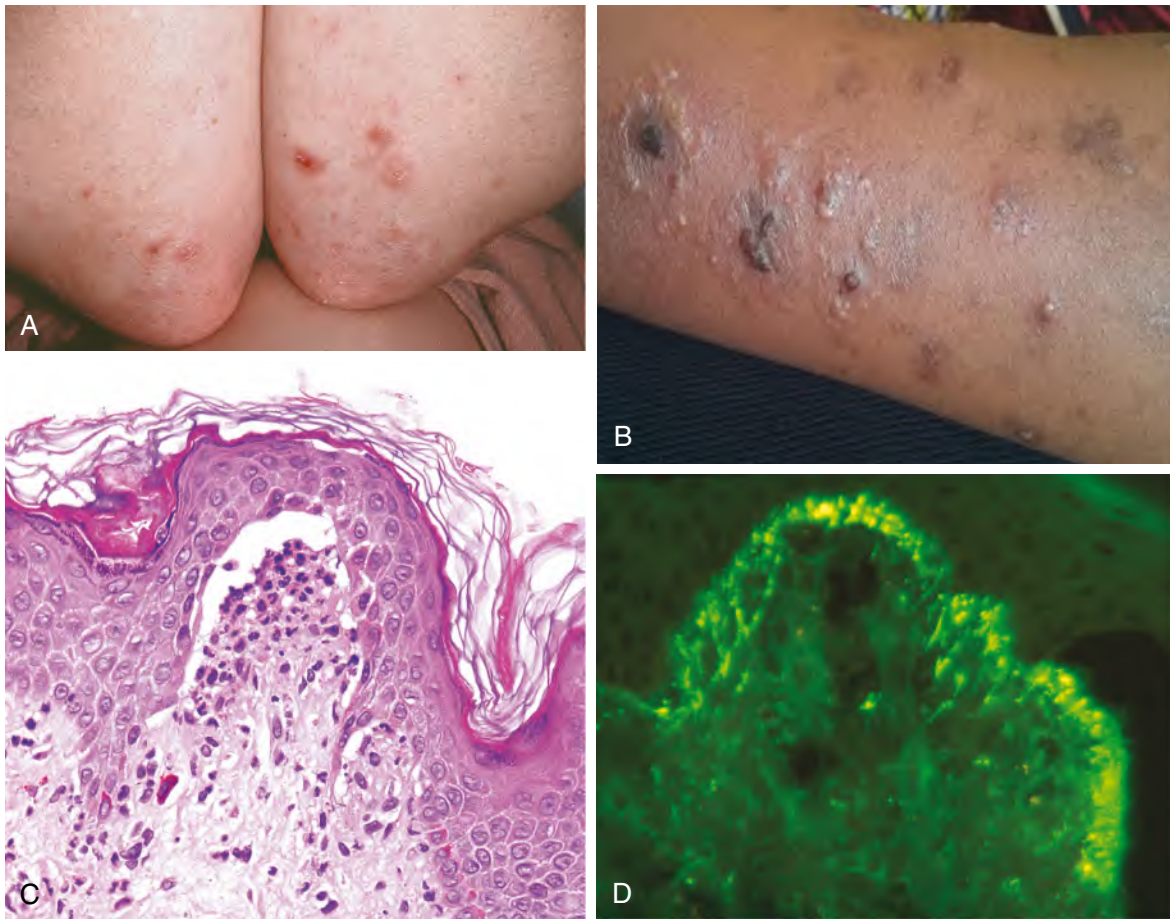


Fig. 25.33 Dermatitis herpetiformis. Lesions consist of intact and eroded (usually scratched) erythematous (A) or hyperpigmented (B) blisters, often grouped (seen here on elbows and arms). (C) Blisters associated with basal cell layer injury, initially caused by accumulation of neutrophils (microabscesses) at the tips of the dermal papillae. (D) Characteristic deposition of IgA autoantibody at the tips of the dermal papillae. (B, From Machona MS, Gupta M, Mudenda V, Ngalamika O. Dermatitis herpetiformis in an African woman. *Pan Afr Med J.* 2018;30:119. D, Courtesy Dr. Victor G. Prieto, Houston, TX.)



Fig. 25.34 Epidermolysis bullosa. (A) Junctional epidermolysis bullosa showing typical erosions in flexural creases. (B) Subepidermal blister at the level of the lamina lucida. There is no associated inflammation.

DISORDERS OF EPIDERMAL APPENDAGES

Acne Vulgaris

Virtually universal in the middle to late teenage years, acne vulgaris affects males and females, with males tending to have more severe disease. It may be induced or exacerbated by drugs (corticosteroids, adrenocorticotrophic hormone, testosterone, gonadotropins, contraceptives, trimethadione, iodides, and bromides), occupational exposures (cutting oils, chlorinated hydrocarbons, and coal tars), and conditions that favor occlusion of sebaceous glands, such as heavy clothing, cosmetics, and tropical climates. Some families seem to be particularly prone to acne, suggesting a hereditary component.

Acne is divided into noninflammatory and inflammatory types, although both types may coexist. Noninflammatory acne may take the form of open and closed comedones.

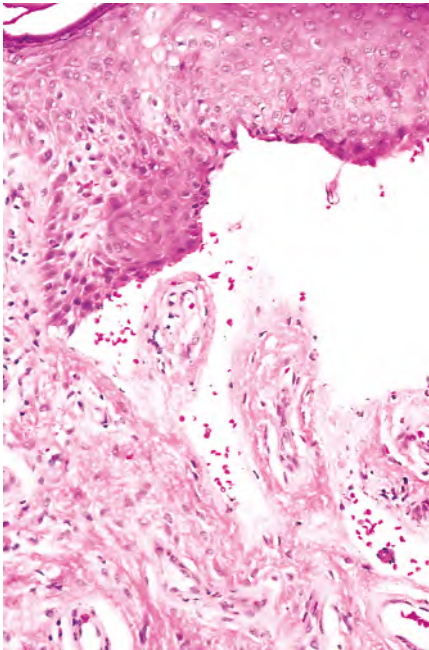


Fig. 25.35 Porphyria. A noninflammatory blister at the dermoepidermal junction; note the seemingly rigid dermal papillae at the base that contain vessels with abnormal walls.

- *Open comedones* are small follicular papules containing a central black keratin plug. This color is the result of oxidation of melanin pigment (not dirt).
- *Closed comedones* are follicular papules without a visible central plug. Because the keratin plug is trapped beneath the epidermal surface, these lesions are potential sources of follicle rupture and inflammation.

Pathogenesis

The pathogenesis of acne is incompletely understood and is likely multifactorial. At least four factors contribute to its development: (1) keratinization of the lower portion of the follicular infundibulum and development of a keratin plug that blocks outflow of sebum to the skin surface; (2) hypertrophy of sebaceous glands during puberty under the influence of androgens; (3) lipase-synthesizing bacteria (*Propionibacterium acnes*) colonizing the upper and mid-portion of the hair follicle, converting lipids within sebum to proinflammatory fatty acids; and (4) secondary inflammation of the involved follicle. Androgens were first implicated in times past when it was noted that young, castrated males generally did not develop the condition. Elimination of *P. acnes* is the rationale for administration of antibiotics to individuals with inflammatory acne. The synthetic vitamin A derivative 13-*cis*-retinoic acid (isotretinoin) brings about remarkable improvement in some cases of severe acne through its strong antisebaceous action.

MORPHOLOGY

Inflammatory acne is marked by erythematous papules, nodules, and pustules (Fig. 25.36A). Severe variants (e.g., **acne conglobata**) result in sinus tract formation and dermal scarring. Depending on the stage of the disease, open or closed comedones, papules, pustules, or deep inflammatory nodules may develop. **Open comedones** have large, wide open (patulous) orifices, whereas the orifices of **closed comedones** are identifiable only microscopically (Fig. 25.36B,C). Variable infiltrates of lymphocytes and macrophages are present in and around affected follicles, and extensive acute inflammation accompanies follicle rupture. Dermal abscesses may form in association with rupture and lead to scarring.

Rosacea

Rosacea is a common disease of middle age and beyond, affecting up to 3% of the US population, with a predilection for females. Four stages are recognized: (1) flushing episodes (pre-rosacea), (2) persistent

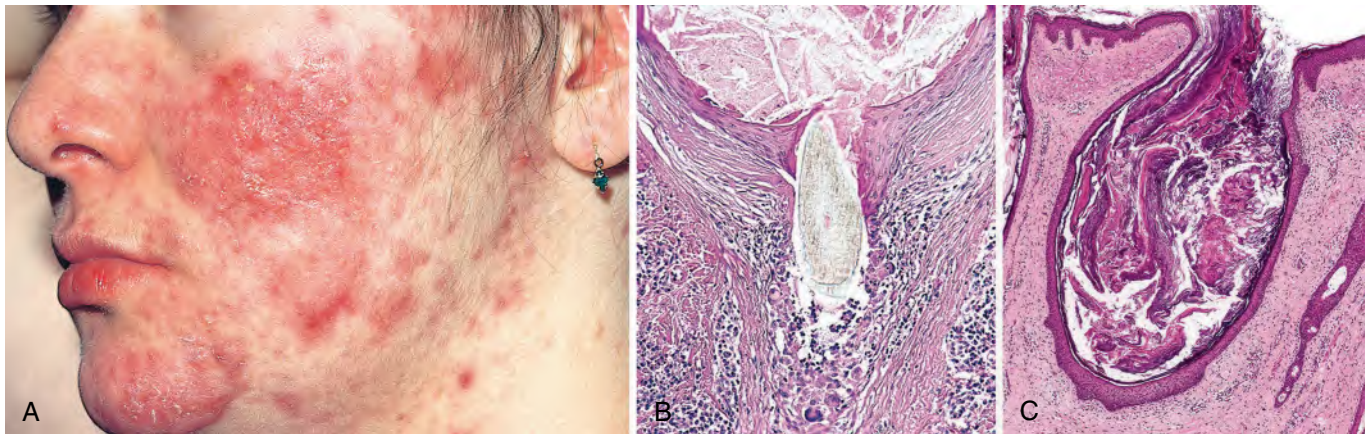


Fig. 25.36 Acne. (A) Inflammatory acne associated with erythematous papules and pustules. (B) A hair shaft pierces the follicular epithelium, eliciting inflammation and fibrosis. (C) An open comedone.

erythema and telangiectasia, (3) pustules and papules, and (4) rhinophyma (permanent thickening of the nasal skin by confluent erythematous papules and prominent follicles).

Pathogenesis

A clue comes from observations showing that individuals with rosacea have high cutaneous levels of the antimicrobial peptide cathelicidin, a mediator of the cutaneous innate immune response. The cathelicidin peptides in affected individuals are qualitatively distinct from those seen in individuals without rosacea because of alternative processing by proteases such as kallikrein 5 (also known as stratum corneum tryptic enzyme). When injected into mice, cathelicidin peptides isolated from patients induce some of the cutaneous changes seen in rosacea, including inflammation and vascular dilation. It has been shown that activation of TLR2 upregulates kallikrein 5 expression in keratinocytes, suggesting that factors that stimulate TLR2 are involved. Several microbial triggers have been proposed, but none are proven.

MORPHOLOGY

Rosacea is characterized by a nonspecific perifollicular infiltrate composed of lymphocytes surrounded by dermal edema and telangiectasia. In the pustular phase, neutrophils may colonize the follicles, and follicle rupture may occur and cause a granulomatous dermal response. The development of **rhinophyma** (red bulbous nose) is associated with hypertrophy of sebaceous glands and follicular plugging by keratotic debris.

PANNICULITIS

Erythema Nodosum and Erythema Induratum

Panniculitis is an inflammatory reaction in the subcutaneous adipose tissue that may preferentially affect (1) the lobules of fat or (2) the connective tissue that separates fat into lobules. Panniculitis often involves the lower legs. *Erythema nodosum* is the most common form and usually has a subacute presentation. A second somewhat distinctive form, *erythema induratum*, also merits brief discussion.

- *Erythema nodosum* presents as poorly defined, exquisitely tender, erythematous plaques and nodules that may be more readily palpated than seen. Its occurrence is often associated with infections (such as by β -hemolytic streptococci, tuberculosis and, less commonly, coccidioidomycosis, histoplasmosis, and leprosy), drug administration (sulfonamides and oral contraceptives), sarcoidosis, inflammatory bowel disease, and certain malignant neoplasms, but many times a cause cannot be identified. Fever and malaise may accompany the cutaneous signs. It is thought to be caused by a delayed hypersensitivity reaction to microbial or drug-related antigens. Immune complexes have been implicated in some cases, but in most instances the pathogenesis is mysterious. Over the course of weeks, lesions usually flatten and become bruise-like, leaving no residual scar, while new lesions continue to develop. Biopsy of a deep wedge of tissue to generously sample the subcutis is usually required for histologic diagnosis.
- *Erythema induratum* is an uncommon type of panniculitis of unknown cause that affects primarily adolescents and menopausal females. Most observers regard it to be a primary vasculitis of deep vessels supplying the fat lobules of the subcutis; the associated vascular compromise leads to fat necrosis and inflammation. *Erythema induratum* presents as an erythematous, slightly tender nodule that usually goes on to ulcerate. Originally considered a hypersensitivity response to tuberculosis, *erythema induratum* today most commonly occurs without an associated underlying disease.

MORPHOLOGY

The histopathology of **erythema nodosum** is distinctive. In early lesions the connective tissue septa are widened by edema, fibrin exudation, and neutrophilic infiltration. Later, infiltration by lymphocytes, macrophages, multinucleated giant cells, and occasional eosinophils is associated with septal fibrosis. Vasculitis is not present. In **erythema induratum**, on the other hand, granulomatous inflammation and zones of caseous necrosis involve the fat lobule. Early lesions show necrotizing vasculitis affecting small- to medium-sized arteries and veins in the deep dermis and subcutis.

Other Types of Panniculitis

Many other types of panniculitis have been described, a few of which merit brief mention.

- *Weber-Christian disease (relapsing febrile nodular panniculitis)* is a rare form of lobular, nonvasculitic panniculitis seen in children and adults. It is marked by crops of erythematous plaques or nodules, predominantly on the lower extremities, created by deep-seated foci of inflammation containing aggregates of foamy macrophages admixed with lymphocytes, neutrophils, and giant cells.
- *Factitial panniculitis* is a form of secondary panniculitis caused by self-inflicted trauma or injection of foreign or toxic substances.
- Rare types of *T-cell lymphoma* home to fat lobules, producing fat necrosis and superimposed inflammation that mimics panniculitis.
- *Systemic lupus erythematosus* may occasionally cause inflammation of the subcutis and an associated panniculitis.

INFECTION

Despite its ample protective mechanisms, exposure of the skin to frequent trauma and the external environment leaves it vulnerable to infection by microorganisms, parasites, and insects. We have already discussed the possible role of bacteria in the pathogenesis of acne, and the dermatoses resulting from viruses are too numerous to list. In the setting of the immunocompromised individual, ordinarily trivial cutaneous infections may become life threatening. Many disorders, such as herpes simplex and herpes zoster, viral exanthems, deep fungal infections, and immune reactions in skin provoked by infectious agents, are discussed in Chapter 8. Here we cover a representative sampling of common infections whose primary clinical manifestations are in the skin.

Verrucae (Warts)

Verrucae are proliferative lesions caused by human papillomaviruses (HPVs). They are common lesions of children and adolescents, although they may be encountered at any age. Transmission of disease usually involves direct contact between individuals or autoinoculation. Verrucae are generally self-limited, regressing spontaneously within 6 months to 2 years.

Pathogenesis

More than 150 types of papillomavirus have been identified, many of them capable of producing warts in humans. The clinical variants of warts are often associated with distinct HPV subtypes. For example, anogenital warts are caused predominantly by HPV types 6 and 11. HPV type 16 has been associated with in situ squamous cell carcinoma of the genitalia and with bowenoid papulosis; the latter is a genital lesion of young adults with the histologic appearance of carcinoma in situ, but which usually regresses spontaneously (see also Chapter 21). The relationship of HPV subtypes 5 and 8 to squamous cell carcinomas in individuals affected by the rare condition epidermodysplasia

verruciformis was mentioned earlier. These patients develop multiple flat warts that contain HPV genomes, some of which progress to carcinoma. Viral typing can be accomplished by either in situ hybridization or polymerase chain reaction.

HPVs that are associated with a high risk of cancer produce E6 proteins that abolish p53 function (Chapter 7). By contrast, HPV subtypes 5 and 8 produce variant E6 proteins that do not affect p53, explaining why these forms of HPV have low oncogenic potential. The E6 proteins of low-risk HPVs interfere with Notch signaling, which is required for the normal maturation of keratinocytes; this effect likely contributes to the epidermal hyperplasia that characterizes warts.

MORPHOLOGY

The classification of verrucae is based largely on appearance and location.

Verruca vulgaris is the most common type of wart. The lesions of verruca vulgaris may occur anywhere but are most often found on the hands, particularly on the dorsal surfaces and periungual areas. On lighter pigmented skin, they typically appear as gray-white to tan, flat to convex, 0.1- to 1-cm papules with a rough, pebblelike surface (Fig. 25.37A), whereas they may appear hyperpigmented on individuals with darker skin (Fig. 25.37B). **Verruca plana**, or **flat wart**, is common on the face or the dorsal surfaces of the hands. The warts are slightly elevated, flat, smooth, tan papules that are generally smaller than verruca vulgaris. **Verruca plantaris** and **verruca palmaris** occur on the soles

and palms, respectively. These rough, scaly lesions may reach 1 to 2 cm in diameter and may be confused with ordinary calluses. **Condyloma acuminatum (venereal wart)** occurs on the penis, female genitalia, urethra, perianal areas, and rectum. Venereal warts appear as soft, tan, cauliflower-like masses that occasionally reach many centimeters in diameter.

Histologic features common to verrucae include epidermal hyperplasia that is often undulant or spire-like in character, termed **verrucous** or **papillomatous epidermal hyperplasia** (Fig. 25.37C), and **cytoplasmic vacuolization (koilocytosis)** involving the more superficial epidermal layers, producing halos of pallor surrounding infected nuclei. Electron microscopy of these zones reveals numerous HPV virions within nuclei. Infected cells may also demonstrate prominent condensed keratohyaline granules and jagged eosinophilic intracytoplasmic keratin aggregates resulting from viral cytopathic effects (Fig. 25.37D). These cellular alterations are not as prominent in condylomas; hence, their diagnosis is based primarily on hyperplastic papillary architecture containing wedge-shaped zones of koilocytosis.

Molluscum Contagiosum

Molluscum contagiosum is a common, self-limited viral disease of the skin caused by a poxvirus. The virus is characteristically brick shaped, has a dumbbell-shaped DNA core, and measures 300 nm in maximal dimension, making it one of the largest pathogenic poxviruses in humans and one of the largest viruses in nature. Infection is usually spread by direct contact, particularly among children and young adults.



Fig. 25.37 Verruca vulgaris. (A) Multiple warts, with characteristic rough, pebblelike surfaces. (B) On more darkly pigmented skin, warts often appear to be hyperpigmented. (C) Microscopically, common warts contain zones of papillary epidermal proliferation that often radiate symmetrically like the points of a crown. (D) At higher power, pallor or halos around nuclei, prominent keratohyalin granules, and related cytopathic changes are seen. (B, From Plastic Surgery Key, <https://plasticsurgerykey.com/27-viral-diseases-of-skin-and-mucosa/>.)

MORPHOLOGY

Multiple lesions may occur on the skin and mucous membranes, with a predilection for the trunk and anogenital areas. Individual lesions are firm, often pruritic, pink to skin-colored, umbilicated papules, generally ranging in diameter from 0.2 to 0.4 cm. Rarely, “giant” lesions occur measuring up to 2 cm in diameter. A curd-like material can be expressed from the central umbilication. Smearing this material onto a glass slide and staining with Giemsa often shows diagnostic molluscum bodies.

On microscopic examination, lesions show cup-like verrucous epidermal hyperplasia. The diagnostically specific structure is the **molluscum body**, a large (up to 35 μm), ellipsoid, homogeneous, cytoplasmic inclusion that is found in the cells of the stratum granulosum and the stratum corneum (Fig. 25.38). In hematoxylin and eosin stains, the inclusions are eosinophilic in the stratum granulosum and acquire a pale blue hue in the stratum corneum. Numerous virions are present within molluscum bodies.

Impetigo

Impetigo is a common superficial bacterial infection of skin. It is highly contagious and is frequently seen in otherwise healthy children as well as occasionally in adults in poor health. The infection usually involves exposed skin, particularly that of the face and hands. Two forms exist, classically referred to as *impetigo contagiosa* and *impetigo bullosa*; they differ from each other simply by the size of the pustules. Over the past several decades a remarkable shift in etiology has been observed. Whereas in the past impetigo contagiosa was almost exclusively caused by group A β -hemolytic streptococci and impetigo bullosa by *Staphylococcus aureus*, both are now usually caused by *S. aureus*.

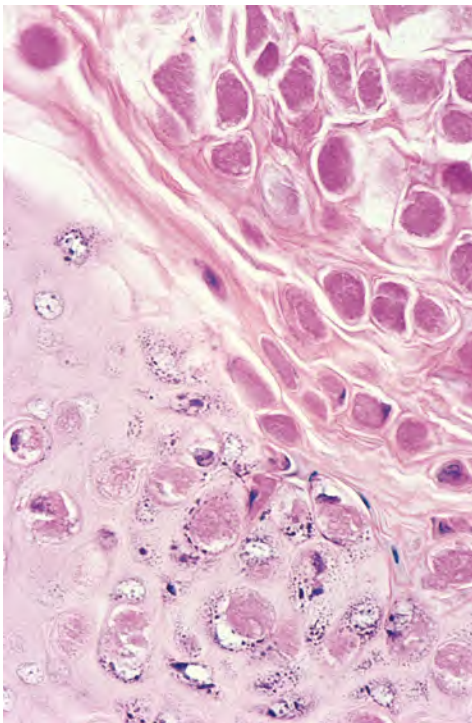


Fig. 25.38 Molluscum contagiosum. A focus of verrucous epidermal hyperplasia contains numerous cells with ellipsoid cytoplasmic inclusions (molluscum bodies) within the stratum granulosum and stratum corneum.

Pathogenesis

Bacteria in the epidermis evoke an innate immune response that causes epidermal injury, leading to local serous exudate and formation of a scale crust (scab). The pathogenesis of blister formation in impetigo is related to bacterial production of a toxin that specifically cleaves desmoglein 1, the protein responsible for cell-to-cell adhesion within the uppermost epidermal layers. Recall that in pemphigus foliaceus, which has a similar plane of blister formation, desmoglein 1 is compromised not by a toxin but by an autoantibody. Because there is virtually no involvement of the dermis, once the bacteria are eliminated the lesions heal without scarring.

MORPHOLOGY

Impetigo presents as an erythematous macule, but multiple small pustules rapidly supervene. As pustules break, shallow erosions form, covered with drying serum, giving the characteristic appearance of **honey-colored crust**. If the crust is not removed, new lesions form about the periphery, and extensive epidermal damage may ensue. A bullous form of impetigo mainly occurs in children.

The characteristic microscopic feature of impetigo is **accumulation of neutrophils beneath the stratum corneum**, often producing a subcorneal pustule containing serum proteins and inflammatory cells. Special stains reveal the presence of bacteria in these foci. Nonspecific, reactive epidermal alterations and superficial dermal inflammation accompany these findings. Rupture of pustules releases serum, neutrophils, and cellular debris, which layer out and dry to form the characteristic crust.

Superficial Fungal Infections

As opposed to deep fungal infections of the skin, where the dermis or subcutis is primarily involved, superficial fungal infections of the skin are confined to the stratum corneum and are caused primarily by dermatophytes. These organisms grow in the soil and on animals and produce diverse lesions with characteristic anatomic distributions:

- *Tinea capitis* usually occurs in children and is only rarely seen in infants and adults. It is a dermatophytosis of the scalp characterized by asymptomatic, patchy skin lesions associated with mild erythema, crust formation, scaling, and frequent hair loss.
- *Tinea barbae* is a dermatophyte infection of the beard area that affects adult males; it is relatively uncommon.
- *Tinea corporis*, on the other hand, is a common superficial fungal infection of skin that affects persons of all ages, but particularly children. Predisposing factors include excessive heat and humidity, exposure to infected animals, and chronic dermatophytosis of the feet or nails. The most common type of tinea corporis is an expanding, round, slightly erythematous plaque with an elevated scaling border (Fig. 25.39A).
- *Tinea cruris* occurs most frequently in the inguinal areas of obese males during warm weather. Heat, friction, and maceration all predispose to its development. The infection usually first appears on the upper inner thighs as moist, red patches with raised scaly borders.
- *Tinea pedis* (athlete's foot) affects 30% to 40% of the population at some time in their lives. There is diffuse erythema and scaling, often initially localized to the web spaces. Most of the inflammatory reaction, however, appears to be the result of bacterial superinfection and is not directly related to the primary dermatophytosis. Spread to (or primary infection of) the nails is referred to as *onychomycosis*. This produces discoloration, thickening, and deformity of the nail plate.
- *Tinea versicolor* usually occurs on the upper trunk and is highly distinctive in appearance. Caused by *Malassezia furfur* (a yeast, not a dermatophyte), the lesions consist of groups of macules of varied size and color with a fine peripheral scale.

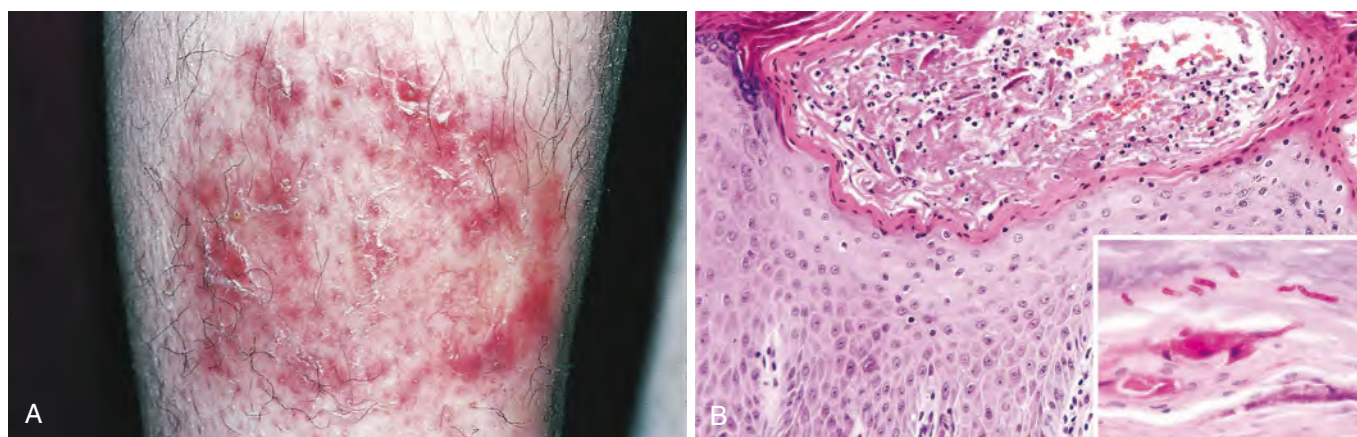


Fig. 25.39 Tinea. (A) Characteristic plaque of tinea corporis. (B) Routine histology shows a mild eczematous (spongiotic) dermatitis and focal neutrophilic abscesses. Periodic acid–Schiff stain (*inset*) reveals deep red hyphae within the stratum corneum.

MORPHOLOGY

The histologic features of dermatophytoses are variable, depending on the properties of the organism, the host response, and the degree of bacterial superinfection. There may be mild eczematous dermatitis associated with intraepidermal neutrophils (Fig. 25.39B). Due to cell walls rich in mucopolysaccharides, fungi stain bright pink to red with periodic acid–Schiff stain. They are found in the anucleate cornified layer of lesional skin, hair, or nails (Fig. 25.39B, *inset*). Culture of material scraped from these areas usually permits identification of the offending species.

SELECTED READINGS

- Alwan W, Nestle FO. Pathogenesis and treatment of psoriasis: exploiting pathophysiological pathways for precision medicine. *Clin Exp Rheumatol*. 2015;33:S2–S6. [Pathogenesis, clinical features, and targeted treatment options are discussed.]
- Antia C, Baquerizo K, Korman A, Bernstein JA, Alikhan A. Urticaria: a comprehensive review: epidemiology, diagnosis, and work-up. *J Am Acad Dermatol*. 2018;79:599–614. [A comprehensive review on urticaria.]
- Boncianni D, Verdelli A, Bonciolini V, et al. Dermatitis herpetiformis: from genetics to the development of skin lesions. *Clin Dev Immunol*. 2012;239691. [Description of factors and disease mechanisms underlying this rare disorder.]
- Breslow A. Prognosis in cutaneous melanoma: tumor thickness as a guide to treatment. *Pathol Annu*. 1980;15:1–22. [Classic article describing the importance of vertical growth of melanoma, which remains the best predictor of outcome in this disease.]
- Cancer Genome Atlas Network. Genomic classification of melanoma. *Cell*. 2015;16:1681–1696. [Comprehensive genomic characterization of melanoma demonstrating categories relevant to targeted treatment.]
- Coenraads PJ. Hand eczema. *N Engl J Med*. 2012;367:1829–1837. [Thorough discussion of the causes and management of a common form of eczema.]
- Drozdowski R, Spaccarelli N, Peters MS, Grant-Kels JM. Dysplastic nevus part I: Historical perspective, classification, and epidemiology. *J Am Acad Dermatol*. 2023;88:1–10. [A comprehensive review on the classification, diagnosis, and historical context of dysplastic nevus.]
- Epstein EH. Basal cell carcinomas: attack of the hedgehog. *Nat Rev Cancer*. 2008;8:743–54. [Epidemiology, clinical presentation, molecular pathogenesis, and novel treatment options are succinctly reviewed.]
- Erstad DJ, Witt RG, Wargo JA. Neoadjuvant therapy for melanoma: new and evolving concepts. *Clin Adv Hematol Oncol*. 2022;20:47–55. [Excellent review on targeted therapies and immunotherapy for melanoma.]
- Maher NG, Scolyer RA, Colebatch AJ. Biology and genetics of acquired and congenital melanocytic naevi. *Pathology*. 2023;55:169–177. [Updated discussion of nevi and their pathogenesis.]
- Moy AP, Pulitzer MP. Aggressive cutaneous lymphomas and their mimics. *Surg Pathol Clin*. 2023;16:361–383. [Summary of cutaneous lymphomas and their manifestations.]
- Pollmann R, Schmidt T, Eming R, et al. Pemphigus: a comprehensive review on pathogenesis, clinical presentation and novel therapeutic approaches. *Clin Rev Allergy Immunol*. 2018;54:1–25. [An update on basic and clinical aspects of pemphigus.]
- Sharma A, Bialynicki-Birula R, Schwatz RA, et al. Lichen planus: an update and review. *Cutis*. 2012;90:17–23. [Review of pathology and clinical features, natural history, and treatment.]
- Volertas S, Schuler CF 4th, Akin C. New insights into clonal mast cell disorders including mastocytosis. *Immunol Allergy Clin North Am*. 2018;38:341–350. [Update on pathogenesis and treatment of mast cell diseases.]
- Zaenglein AL. Acne vulgaris. *New Engl J Med*. 2018;379:1343–1352. [Excellent report on how to assess and treat acne.]